PRODUCT REQUIREMENTS DOCUMENT · V1.0

MAGNETOPE

A magnetic rage-climb for the open web

One button flips your polarity - stick, launch, snap - up a colossal junk mountain where every fall is slapstick, every retry is instant, and every fail is a clip. Engineered to clear Poki's bar: original, wholesome, web-first, and playable in under six seconds.

Magnetope Studio

Solo developer production · Three.js + Rapier + Web Audio

OCTOBER 2026 · TARGET PLATFORM: POKI.COM

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1 Executive Summary

Poki is the largest curated web gaming platform in the world: over 100 million monthly players, more than one billion gameplays per month, and the number one web gaming destination in over 100 countries. Unlike open app stores, Poki is aggressively curated, which is precisely why it is worth targeting. The platform's own developer documentation states that a high-performing game reaches more than five million monthly players, and its 2026 State of Web Gaming study reports a browser games market of roughly 8.01 billion US dollars, growing toward 9.07 billion by 2030. The audience is majority mobile, global, and famously impatient: players abandon a game that takes more than ten seconds to load, and the first session carries almost all of the lifetime value.

At the same time, a flood of low-effort, AI-generated games is degrading trust across every game platform. Steam Next Fest has been publicly criticized for being overrun with AI content, Valve has rejected games over AI-asset copyright uncertainty, and Poki's own guidelines now include an explicit section on AI-heavy production: **"AI should empower, not shortcut."** Games built from template assets, stock loops, and cloned mechanics are rejected on originality grounds even when they function correctly. This creates a genuine opening for a solo developer with real craft skills: the bar for acceptance has risen, and most submitters cannot clear it.

MAGNETOPE is the answer this document proposes. It is a 3D low-poly physics rage-climber in which the player **is a magnet**: one button flips the player's polarity between attract and repel, turning a colossal mountain of scrap into a vertical slingshot puzzle. Falling is slapstick, not gruesome; retrying takes one tap; every failure produces a shareable comedy moment. The concept occupies a validated but unclaimed quadrant on the web: the emotional loop of Getting Over It, the level structure and instant restarts of Drive Mad, and the vertical junk-tower fantasy of Only Up - none of which exists on Poki today in a premium, family-friendly, web-native form.

100M+	≤ 4 MB	~3 min	$500–$3,000
monthly players on Poki across 100+ countries	initial download target vs Poki's 5 MB ceiling	core loop length tuned to session-0 behavior	realistic monthly revenue band for a first web hit

The creator's skill stack is the moat. Sound engineering means every surface in the game has a material-coded voice and the music adapts to altitude - no stock loops anywhere in the build. VFX and motion graphics mean magnetic field lines, slow-motion falls, and debris avalanches that read as premium rather than generated. Machine learning runs strictly as development tooling - playtest bots that verify every level is completable and quantify difficulty - while the shipped game stays tiny. The stack is Three.js, TypeScript, and Rapier (WASM physics), which load in two to six seconds where Unity WebGL commonly takes eight to thirty.

The plan asks for one developer, six months, and a budget of effectively zero. Monetization is entirely Poki's ad system: midroll ads at natural stopping points, optional rewarded videos, and Gamebar display revenue from portrait play. This document specifies every domain of the build in production-grade detail - design, art, animation, VFX, audio, engineering, ML, UX, monetization, QA, launch, and analytics - so that development can begin the day it is approved and so that the finished game clears Poki's final review on the first submission.

2 Research Foundations: The Poki Playbook

2.1 The Platform in Numbers

Poki reports 100 million monthly players and over one billion gameplays per month, sustained by a community of roughly 500 developers. The platform's developer portal markets itself on three promises: instant play (no downloads, no login), a global audience across web and mobile browsers, and full-stack support - Poki handles hosting, marketing, QA, brand safety, and ad sales, while the developer keeps making games. Testimonials on the portal from Unept (Level Devil), Fancade (Drive Mad), Madbox, Blumgi, SYBO (Subway Surfers) and Deer Cat Games consistently emphasize the same thing: a small, well-made web game on Poki can outperform the same game on any other platform the developer has tried.

The market context supports the timing. The Business Research Company values browser games at 7.81 billion dollars in 2025, 8.01 billion in 2026, and projects 9.07 billion by 2030. Poki's own 2026 State of Web Gaming study - built from 2,000 surveyed players and 400 developers - found that 92% of consumers describe HTML5 games positively, that 27% of web gamers spend over 50 dollars on gaming per period, and that 53% of developers plan to bring mobile games to the browser. Separate industry forecasts put the HTML5 games segment at roughly 6.0 billion dollars in 2026 growing to 10.5 billion by 2035. Web is not a legacy channel; it is a compounding one.

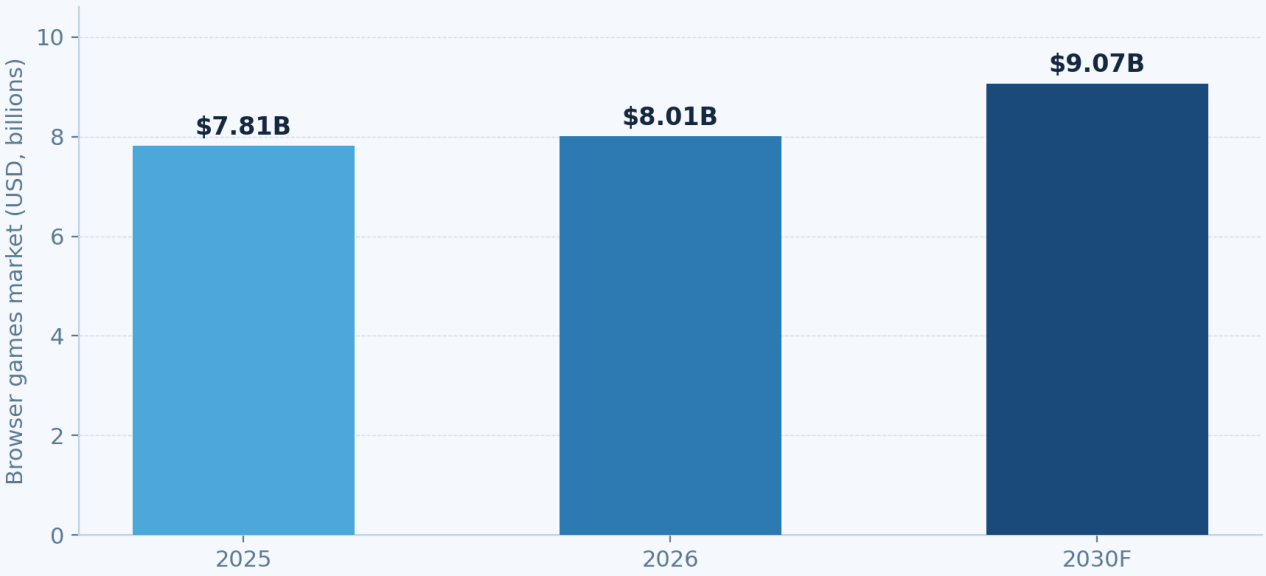


Figure 2.1 - Browser games market value (The Business Research Company, 2025-2026, 2030 forecast)

2.2 What Poki Hard-Requires (Non-Negotiable)

Poki's Requirements page defines a hard technical floor that QA checks before any game can go live, and every item below is quoted or paraphrased from the official documentation. These are pass/fail gates, not preferences, and MAGNETOPE's engineering chapter is written to satisfy each one by construction rather than

by late-stage patching.

Poki hard requirement	Engineering consequence for MAGNETOPE
Desktop + mobile + tablet support; full-screen coverage; mobile control scheme forced on tablets	Responsive 16:9 canvas, touch-first control layer, UI safe areas, auto-detection of tablet form factor
Scale proportionally to 640×360, 836×470, or 1031×580	Fixed logical resolution with letterbox + dynamic resolution scaling tiers
Incognito support (localStorage restricted)	All storage wrapped in try/catch with in-memory fallback; game fully playable with zero persistence
No external requests - fonts, assets, libraries all bundled	Zero-runtime-fetch build; Google Fonts rejected; self-hosted subset font in bundle; CSP-safe by design
No splash screens, outgoing links, or non-Poki ads	Studio logo appears only on the loading bar; every screen is gameplay-first
Ad-blocker compatibility - game must stay playable with blockers active	No ad-logic dependencies in the game loop; Poki SDK ad calls are advisory, never blocking
SDK events: gameplayStart() on first input, gameplayStop() on every interruption, no double-fires, no events during ads	Single AudioBus-style event broker wraps all state transitions; automated event trace tests in CI
Small file size; initial download ≤ 5 MB, total ≤ 8 MB; players leave after ~10 s	Progressive loading architecture: 3.85 MB initial payload, biome chunks streamed after session start

Table 2.1 - Poki hard requirements and their engineering consequences

2.3 Why Games Get Rejected

Poki documents its rejection logic with unusual candor, and the pattern matters more than any single rule. Two categories are rejected immediately: **misused IP** (anything that borrows recognizable characters, names, or trade dress) and **adult themes** - graphic violence including open wounds and visible body fluids, sexual content, sustained scary themes, gambling, alcohol, tobacco, bullying, stereotypes, and sexism. Poki is a playground that includes children, and its advertising partners must be comfortable placing brands next to every game. This is why the classic web physics-comedy lineage of Happy Wheels - which is built on gore - cannot simply be ported to Poki, and why MAGNETOPE's fail states are designed as pure slapstick: tumbling, bouncing, and comedic collapse, never injury.

Beyond the hard categories, the most common quality declines are **buggy or broken gameplay, excessive load size or poor performance, confusing UX, and poor or inconsistent graphics**. A subtle but decisive rule follows: games that feel **too derivative** may be declined even when they work perfectly, and when a genre or trend becomes crowded, Poki actively declines additional similar submissions because overlapping games fragment the audience and drag each other down. Saturated or high-risk categories as of October 2026 include generic 2D platformers, .io arena clones, dress-up variants, merge and idle management, block puzzles, stickman fighting, and endless runners.

Poki assesses originality across six explicit dimensions, and this table - adapted from the official Content & Player Safety guide - becomes the design contract that Chapter 3 defends point by point:

Dimension	Clone (rejected)	Inspired (accepted)
Art style	Near-identical visuals, colors, themes	Reimagined style or a completely different theme
Core mechanics	Copied loop, goals, and inputs	A unique mechanic, twist, or genre blend
Progression & economy	Same structure, pacing, monetization	Original level flow and progression logic
UI, icons, audio	Replicated layout, colors, or audio cues	A distinct interface with custom assets
Characters	Same designs, animations, silhouettes	Original characters, no IP infringement
Name & thumbnail	Intentionally similar to ride search terms	Presentation focused on your own content

Table 2.2 - Poki's clone-versus-inspired assessment (official dimensions)

2.4 The AI Production Bar

Poki's Working-with-AI guidance is the single most important document for this project, because the developer's stated plan is to build with an AI subscription and personal craft skills. The golden rule reads: **AI should empower, not shortcut**. Prototyping with templates, inspiration, and AI tools is normal; but before a game enters player testing it must reflect the creator's unique perspective and show meaningful improvement beyond the base tooling. The more non-original material a game leans on, the higher the risk it will not be accepted - even if it functions well. Ownership matters too: the developer must be able to show how and which tools were used, with prompts and iterations on request, because purely AI-created work can create IP ownership problems.

The guidance also specifies a per-craft do-and-don't table that maps exactly onto this project's workflow: use AI for concepts and then heavily edit into one cohesive direction; never drop raw output in as-is; never ship untested generated levels - every level must be hand-balanced; use AI for code boilerplate while a human owns testing and optimization; keep every visual watermark-free and prompt-text-free. MAGNETOPE institutionalizes these rules as a documented asset pipeline in Chapters 6, 11, and 16, including a kept prompt-and-iteration log that can be shown to Poki on request. Where the average AI-submitted game fails this bar, this project treats the bar as a design constraint from day one.

> *Experimenting with templates, inspiration, and AI tools while prototyping is normal. But before your game enters a player fit or web fit test, it should reflect your unique perspective and meaningful improvements beyond the base tooling.*

2.5 The Engagement Doctrine

Poki's engagement guidance is unusually specific and effectively prescribes the product's shape. Web players have shorter sessions and return less often than mobile players, so the first session carries most of the weight. Quick, satisfying, repeatable loops of around three minutes perform best, and roughly one hour of total content is the right scope - sprawling mobile-style progression systems do not translate, and memory limits how much

content fits in a small build anyway. Games where the player can constantly perform an action outperform games with waiting and downtime. The player should be congratulated at every milestone; difficulty must start easy and ramp gradually; and both touch and keyboard navigation must work through every menu.

Onboarding guidance is equally blunt: skip the menu, skip splash screens and level selects, and put first-time players straight into gameplay in a safe beginner environment where failure is cheap. Tutorials must be visual rather than textual, because walls of text both intimidate and exclude the non-English-speaking half of the audience. Progressive loading - downloading only the menu, tutorial, and first stage, then fetching the rest in the background - is called out as one of the most important techniques for success on the web. Every one of these directives is reflected in MAGNETOPE's session-zero design in Chapter 5 and the chunked build in Chapter 10.

2.6 Design Mandates Derived From the Research

1. Build for the first three minutes first: instant load, no menu, visual tutorial, first clear inside 60 seconds.
2. Keep every promise tiny and repeatable: 30-90 second levels, one-tap retries, constant action with zero downtime.
3. Stay wholesome by construction: slapstick physics comedy, zero gore, zero chat, zero data collection.
4. Own a mechanic nobody on the web owns, and defend it across all six of Poki's clone-assessment dimensions.
5. Respect the byte budget as a design constraint: ≤ 4 MB initial, ≤ 8 MB total, procedural audio over canned music.
6. Use AI as documented labor (concepts, boilerplate, drafts) and hand-craft everything players actually see, hear, or feel.
7. Engineer SDK correctness and incognito safety from the first commit, not as a pre-submission patch.
8. Ship with room to grow: content hooks and a one-year update roadmap shared at submission.

3 Game Concept: MAGNETOPE

3.1 Vision

One-liner: MAGNETOPE is a 3D low-poly physics rage-climber where you are a magnet - one button flips your polarity between stick and repel, turning a colossal junk mountain into a vertical slingshot puzzle where every fall is slapstick, every retry is instant, and every failure is a clip. The fantasy is simple and physical: the player should feel like a fridge magnet that refuses to stay on the ground. Momentum is the real character; the mountain is the real antagonist.

The emotional target is the comedy of ambition. Drive Mad and Level Devil have proven on Poki that teenagers will happily replay a thirty-second gauntlet forty times if the failure is funny, the retry is instant, and the win feels earned. Getting Over It proved that losing real progress produces a specific, watchable, shareable kind of rage. MAGNETOPE fuses those energies while removing the cruelty: falls cost altitude and dignity, not

an entire evening, and the world physically laughs with you - debris avalanches, bouncing girders, a washing machine that boings you off a rooftop.

3.2 Design Pillars

Pillar	Meaning	What it forbids
1. One button, infinite depth	The entire verb set is stick, flip, snap; mastery lives in timing and momentum	Combo buttons, mode switches, tutorial text
2. Fail funny, retry fast	Every failure is slapstick and physical; restart is one tap under 400 ms	Gore, load screens on retry, progress-gating punishment
3. The mountain is the level design	Every climbable object is a readable prop with known material behavior	Abstract floating geometry, unclear surfaces
4. Sound is the soul	Material-coded audio world, adaptive score, zero stock loops	Generic looped music, placeholder SFX
5. Built to grow	Biomes, ghosts, and cosmetic systems structured for year-one content drops	Hard-coded one-shot content, dead-end progression

Table 3.1 - Five design pillars and what each one forbids

3.3 Player, Tone, and Comedy

The primary persona is the 13-18-year-old web gamer: plays in a browser tab on a school Chromebook or a mid-tier Android phone, watches Poki-adjacent content on TikTok and YouTube, discovers games through thumbnails and friends, and values skill expression and shareable fail moments above narrative depth. This player does not read; this player drops in, dies, laughs, retries, and - if the game is good - sends the clip to a friend. The secondary persona is the older casual who played Flash physics games in the 2010s and recognizes the genre's pleasure instantly.

Tone is Saturday-morning-cartoon slapstick crossed with a Saturday-morning yard sale. The world is a monument to abandoned objects - fridges, shopping carts, satellite dishes, gym lockers - stacked into impossible geography. Humor comes from physics truth: a filing cabinet that wobbles before it topples; a trampoline-styled mattress that launches you exactly one meter too far; a magnet-headed robot slowly sliding down a refrigerator door with a rising squeak. There is no dialogue beyond environmental signage, no narrator, and no gore: MAGNO is a robot, so every fail is a tumble, a bounce, and a spring-loaded pop back upright.

3.4 Positioning: The Unclaimed Quadrant

The concept deliberately occupies the intersection of three proven lineages that no premium web game currently combines. **Getting Over It** owns rage-climbing but exists on CrazyGames as a premium-feeling 2D entry with a single unstructured mountain; it is not on Poki. **Drive Mad** - one of Poki's highest-rated games at 4.3 stars with 200+ levels - owns simple-controls-plus-rage in short structured levels, but is a driving game with no climbing, no character comedy, and no physics-body humor. **Only Up** owns vertical junk-tower fantasy and went violently viral on TikTok in 2023, but its web copies are low-effort clones confined to third-tier portals,

and the premium original never shipped to Poki. The quadrant structured + premium + physical-comedy + vertical is empty on the platform, and Poki's own oversaturation policy rewards exactly this kind of diversification.

Named-neighbor scan (October 2026, Poki live catalog and web search): Stickman Hook owns physics swinging; Basket Random and Soccer Random own ragdoll sports; Monkey Mart owns idle retail; Blumgi Slime owns friendly 2D platforming; Level Devil owns troll-platforming; Escape Road owns driving escapes; Hide and Paint owns novel casual hybrids. No Poki game features a polarity-flipping magnet mechanic, and the only magnet-adjacent web games found anywhere - Teslagrad, Super Magbot, Magnetic By Nature, and a 2D puzzle called Magnetic Escape - are 2D precision puzzle-platformers on other platforms, none of them ragdoll climbers, none of them 3D, none of them comedy. The mechanic is proven fun in a different genre and unclaimed in this one: the definition of inspired, not imitated.

3.5 Narrative: Wordless, Environmental, Wholesome

Centuries ago, something magnetically special happened at one spot on Earth. Loose metal from the whole world drifted there over generations and compacted into a mountain - the Magnetope. MAGNO, a small scrap-golem with a horseshoe-magnet head and exactly one eye, lost its power core in an avalanche and wakes at the mountain's base with one goal: climb to the peak and get its heart back. The story is told entirely through environment: faded festival banners in the Scrapyard biome, a suburb's worth of lawn mowers and mailboxes in Suburbia, neon signage in Metropolis, a giant rusted playground in the fourth biome, and finally the thin-air silence of Stratosphere, where the core glows like a lighthouse. The ending is a sixty-second wordless victory: the core clicks into MAGNO's chest, the magnet head's field lines bloom outward across the sky, and the camera pulls back to reveal the whole absurd mountain one last time.

3.6 Naming Strategy

Working title: MAGNETOPE - a portmanteau of magnet and -ope (as in envelope, slope, and telescope) that names the mountain itself. The word is unique: a search across Poki, CrazyGames, itch.io, and general web search in October 2026 returns zero game-title collisions, which satisfies Poki's name-and-thumbnail dimension (presentation focused on the game's own content, not riding search terms). It is short, pronounceable in every launch locale, reads well in uppercase on a thumbnail, and leaves room for sequel naming (MAGNETOPE 2: The Descent). Alternatives held in reserve for A/B testing during soft release: FLUX & TUMBLE (warm, comedic, two words), POLESTAR (premium but colder), and SCRAP SUMMIT (descriptive but generic - held only as a control). Character name MAGNO is similarly collision-free and carries in three syllables.

3.7 Clone-Defense Matrix

This table is the formal defense against the "this already exists" rejection, mapped to Poki's six official assessment dimensions from Table 2.2. It is written to be shown to the Poki team at submission time alongside the research that produced it.

Dimension	MAGNETOPE's position
Art style	Original low-poly diorama direction: flat-shaded junk-mountain biomes, warm-vs-cool lighting story, toy-scale props. No existing climber on any platform uses this treatment.
Core mechanics	Polarity-flip momentum climbing (stick-repel slingshot + airborne snap). Not present in any web physics-climber; magnet mechanics elsewhere are 2D puzzle-platformers with opposite design goals (precision, not momentum comedy).
Progression & economy	Structured 30-90 s vertical gauntlet levels with 3-star ratings, speedrun ghosts, and a single Scrap currency - versus Getting Over It's single unstructured mountain and Only Up's checkpointless sprawl.
UI, icons, audio	Custom minimal HUD (altitude + charge), custom iconography, and a fully material-coded procedural audio world - no shared asset packs, no stock loops.
Characters	MAGNO: original robot silhouette (round-bottom ragdoll, horseshoe-magnet head, single eye) designed for merchandising-scale readability. Zero IP adjacency to any published character.
Name & thumbnail	Ownable coined word with zero title collisions; thumbnail shows MAGNO mid-flip against the tower - no search-term riding.

Table 3.2 - MAGNETOPE versus Poki's six clone-assessment dimensions

4 The Signature Mechanic: Polarity Flip

4.1 The Three Verbs

The entire control surface is three verbs. **STICK**: whenever any part of MAGNO's body contacts a ferromagnetic surface while attract mode is active, a magnetic joint forms instantly and MAGNO adheres - hanging, dangling, or rolling in place on walls and ceilings. **FLIP**: the single player-issued input reverses polarity. If attached, the joint inverts into a repulsor, launching MAGNO away from the surface along its contact normal with a force proportional to charge held; if airborne, flip arms the snap. **SNAP**: while airborne in attract mode, the nearest ferromagnetic surface inside snap radius is highlighted, and releasing flip (or pressing it again, configuration-dependent) pulls MAGNO toward it with a spring force that converts cleanly into orbital momentum. Chaining flip-snap-flip turns the tower into a vertical slingshot course.

Momentum is sacred. The physics model deliberately never zeroes velocity on attach; instead, surface friction bleeds it over 250 milliseconds, so a fast arrival can convert into a wall-crawl or a pendulum swing around a girder's edge. This single choice - conservation over reset - is what makes the mechanic learnable in ten seconds and still improving after ten hours, and it is the difference between this design and a teleporting grapple.

4.2 The Force Model

Parameter	Baseline	Feel intent
Attach snap force	1,800 N ramp over 60 ms	Instant-feeling grab without teleport pop

Parameter	Baseline	Feel intent
Attach friction ramp	0 → μ 0.9 over 250 ms	Momentum converts to slide/crawl instead of dying
Repel impulse (full charge)	14 m/s exit velocity along normal + 30% input tilt	Predictable slingshot players can aim
Repel impulse (tap)	6 m/s	Fine repositioning without full commitment
Snap radius	4.5 m sphere, 15° cone preference toward camera-forward	Forgiving but not magnetic-lock-on
Snap spring constant	k = 240, damping ζ = 0.55	Pull feels like a magnet, lands like a catch
Charge time (hold to power flip)	0.9 s to full, 3-level meter with audio ramp	Risk-reward: holding while dangling is exposure
Air control	±2.4 m/s^2 tilt authority, never cancels gravity	Steering, not flying
Coyote attach window	120 ms after leaving contact	Borderline flips still feel fair
Input buffer	140 ms	Late flip during recovery still registers

Table 4.1 - Physics feel targets for the magnet controller (tuning baselines, not final)

4.3 Material Rules

Every surface in the game declares exactly one material family, and the material grid is short enough to memorize visually in one biome. **Ferromagnetic** surfaces (steel girders, fridges, cars, lockers, filing cabinets, satellite dishes) participate in stick, flip, and snap at full strength. **Weak-metal** surfaces (aluminum gutters, thin sheet metal, cans) hold stick for a degrading 1.2 seconds - long enough to plan, short enough to panic - and are visually thinner and rustier. **Non-metallic** surfaces (wood, rubber, glass, plastic, fabric) never hold stick: MAGNO bounces, slides, or rolls off, and each has its own restitution (mattress 0.75, rubber 0.65, wood 0.3, glass 0.1 with a breakable flag). **Live-ferrous** props (springs, fans, elevator counterweights) are dynamic bodies: MAGNO's field drags them, they drag back, and loose-junk chain reactions emerge from the simulation rather than being scripted.

4.4 Input Map

Action	Keyboard	Touch	Mouse-only (accessibility)
Move / aim tilt	WASD or arrows	Left-half virtual stick (drag anywhere)	Cursor position relative to MAGNO
Flip (hold to charge)	Space / J / left-click	Right-half big button	Left click
Pause	ESC or P	Top-right icon + hardware back	Right click
Menu confirm / navigate	Space, Enter, arrows	Tap	Click
Retry (after fall)	Space / R	Anywhere on screen	Any click

Table 4.2 - Control schemes across devices (all remappable)

4.5 Anti-Frustration Grammar

Rage is the product, but unfairness is poison, so the mechanic obeys a strict fairness grammar. Falls never damage; they only cost altitude and time. The camera pulls back automatically during long falls so the player can read the landing zone, and a soft auto-align rotates MAGNO's snap cone toward the nearest safe hold during the final 300 milliseconds of a fall - an assist that defaults on, is tunable, and can be disabled by expert players for score-boost purposes. Every fall longer than twelve meters triggers the slow-mo system, guaranteeing that even punishment feels like a produced moment. Retry state is captured the instant a run starts, so restart rebuilds the level in under 400 milliseconds with zero loading screens.

4.6 The Skill Ceiling

1. Novice (minutes 1-5): stick, tap-flip, land, repeat. Clearing a level at all is the win. Fall comedy carries the session.
2. Intermediate (hours 1-3): hold-charged flips for distance, snap-chaining across two surfaces, wall-crawls to skip low routes, weak-metal timing.
3. Expert (hours 3+): momentum conservation off dynamic props - bouncing off a falling girder mid-air into a snap; charge-cutting (releasing flip during the 60 ms ramp for mid-power exits); route optimization against ghosts; no-assist runs for leaderboard-grade times.

4.7 State Machine (Reference)

```
enum MagnoState { AIRBORNE, ATTACHED, CHARGING, REPelled, RECOVERING }

onContact(surface, contactNormal, velocity):
if state == AIRBORNE and attract and surface.isFerromagnetic:
attach(surface, contactNormal) // joint + friction ramp 250ms
sfx.attach(surface.material) // material-coded thunk
vfx.sparkBurst(contactPoint, surface.material)

onFlipPressed():
if state == ATTACHED: enter(CHARGING) // audio ramp, meter fills
if state == AIRBORNE: armSnap() // highlight best target

onFlipReleased(heldMs):
power = chargeCurve(heldMs) // tap 6 m/s .. full 14 m/s
impulse = surfaceNormal * power + tilt * 0.30 * power
detach(); applyImpulse(impulse)
sfx.repel(power); vfx.fieldLinePulse() // polarity flash + shake

onFall(heightMeters, impact):
if heightMeters > 12 or impactSpeed > 14: engageSlowMo()
camera.widen(); audio.fallWhistle(pitchBy(heightMeters))
ragdoll.goLimp(); await rest -> RECOVERING -> getUp()
```

5 Game Design Domain

5.1 The Core Loop

The loop is a closed six-step cycle engineered to complete in under three minutes and to leave the player mid-loop when any interruption - ad, tab switch, bus arrival - occurs, which is the web-session survival strategy Poki's engagement doctrine demands. Load dumps the player directly into gameplay; climbing is continuous action with zero downtime; failing is content rather than punishment; retrying costs one tap and 400 milliseconds; scoring pays out three currencies of achievement (stars, speed, Scrap); and unlocking happens at the natural stop where a commercial break can cleanly sit. The diagram below is the canonical reference for every level and system decision in this chapter.

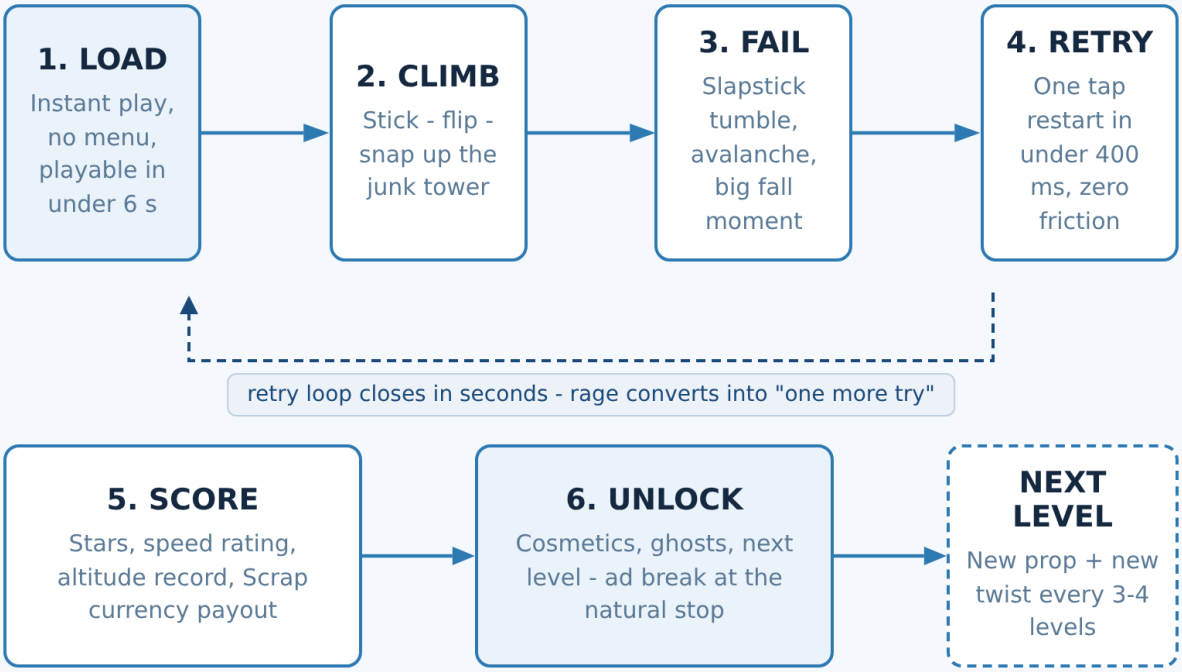


Figure 5.1 - MAGNETOPE core loop; the retry arm closes in seconds so rage converts into one more try

5.2 Level Anatomy

Every level is a vertical gauntlet of 30-90 seconds for a competent first clear, built from four repeating zones. The **base camp** (first 5 seconds) is always safe: a wide platform, one obvious first stick-and-flip, zero threats - this is where session-zero players learn the level's dialect. The **twist** introduces the level's one new idea (a fan that pushes mid-flight, a crane hook that swings, weak-metal shelves that start shedding) with a guaranteed-safe practice encounter. The **gauntlet** combines the twist with the biome's established vocabulary at rising intensity, and is tuned to produce exactly two to four failures for a median player on first visit. The **summit** is the final 10 meters: always visible from the base, always marked by a glowing collection beacon, and always one satisfying charged flip away. Length is a hard budget: levels that playtest longer than 90 seconds get split, no exceptions, because the three-minute loop 容纳 two retries plus a clear.

5.3 Biome Progression

Biome	Setting & props	Mechanic curriculum	Palette & mood
1. Scrapyard	Crushed cars, fridge walls, girder stacks, tire piles, chain fences	Stick, tap-flip, hold-charge, snap; weak-metal introduced at L4; dynamic props (loose girders) at L8	Rust orange + teal shadows, late-afternoon sun, warm dust haze
2. Suburbia	Tool sheds, gutters, mailboxes, satellite dishes, clotheslines, trampolines	Swinging from dynamic anchors; bounce materials; moving surfaces (garage doors)	Mint cream + coral accents, morning light, clean shadows
3. Metropolis	Scaffolding, AC units, billboards, window-washer rigs, crane hooks	Timed snap-chains; wind gusts; weak-metal walls under time pressure; elevator counterweights	Cool blue + neon magenta signage, dusk, deepening fog
4. Playground of Giants	Rusted mega-slide, ferris-wheel spokes, swing-set towers, ball pits	Full momentum play: dynamic-prop launches, orbital swings around spokes, riskier shortcuts	Faded candy pink + sun-bleached yellow, overcast glare
5. Stratosphere	Stratospheric junk: airplane fuselage, satellites, telescope dishes, the Core	Mastery test combining all verbs at low gravity with strong wind; no new tutorials, pure execution	Indigo + starlight, thin air, aurora field lines

Table 5.1 - The five launch biomes (14 levels each, 70 total)

5.4 Difficulty Architecture

Difficulty is specified numerically as a **median fall rate per attempt** - the percentage of level entries that end in a fall rather than a clear - because it is directly measurable by the ML playtest bots of Chapter 11 before a human ever touches the build. The target curve ramps from roughly 6% at level 1 to 68% at level 70, with each biome's opening levels stepping down 5-8 points to grant a breather after the previous biome's climax. The rage band - the range where failure feels funny rather than unfair - sits between 18% and 35% for mid-game levels; above 40% the level is flagged for softening, and below 12% it is flagged as filler and gets a twist sharpened. Spikes are deliberately built in: roughly one in five levels is a **conflict level** that runs 10-15 points above the local trend, because those are the levels that become clips, TikTok captions, and schoolyard legend. The curve below is the shipping contract; the bots enforce it per level and per biome before every content milestone.

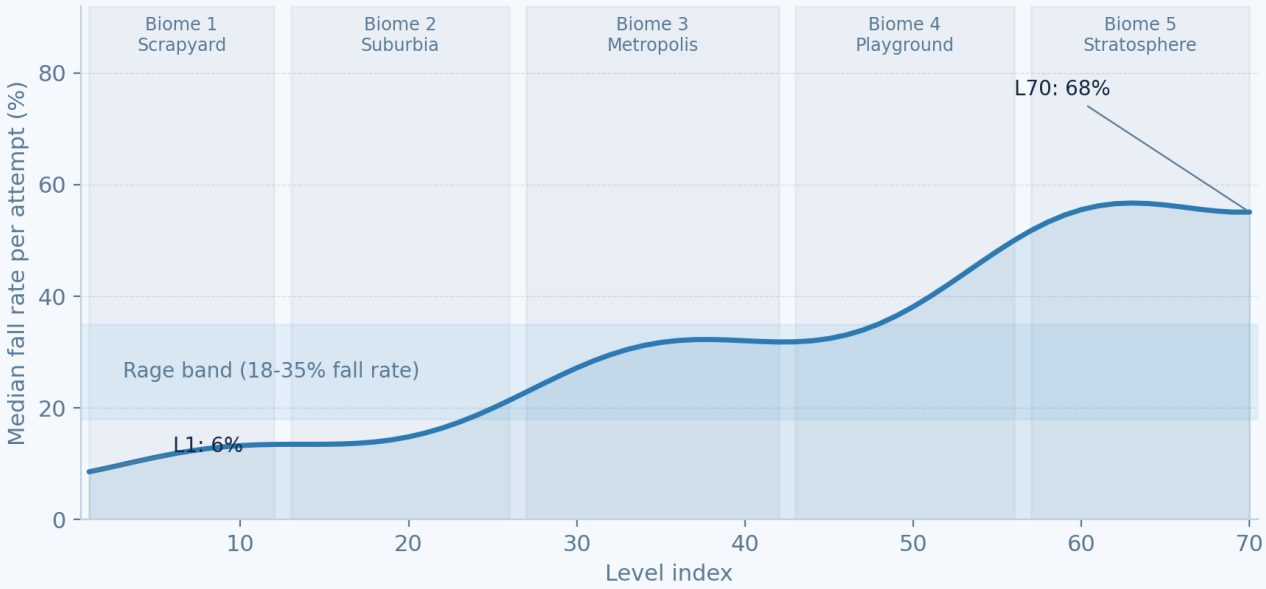


Figure 5.2 - Target median fall-rate curve across the 70 launch levels, with biome bands and the rage zone

5.5 Progression Systems

Three parallel progression tracks keep three kinds of player moving forward. **Stars**: every level awards up to three - one for clearing, one for clearing without a fall beyond the safety line (the no-fall run), and one for beating the par time shown on the level card. Stars gate biome unlocks at soft thresholds (biome 2 at 15 stars, biome 3 at 34, biome 4 at 57, biome 5 at 80), so a player who struggles can still advance by clearing optional challenges, while completionists have 210 stars to chase. **Speedrun ghosts**: every level's par time ships with a built-in developer ghost; beating par unlocks a personal ghost, and Chapter 15's AUDS-based ghost exchange makes friend ghosts and regional top-10 ghosts appear in-world. **Scrap**: the single meta currency, earned 10-50 per level on first clear and 2-8 on replays, spent purely on cosmetics (Chapter 13) so progression never gates content behind grind - a deliberate Poki-doctrine choice that keeps the natural reward path free.

5.6 Fail States and the Comedy System

Failure is a designed content category, not an absence of success. Physically, every fall switches MAGNO to limp ragdoll with exaggerated joint looseness; the body tumbles, bounces with material-appropriate boings, drags loose junk into a small avalanche, and settles. The recovery system then plays a procedural get-up - a spring-loaded pop upright that reads as dignified stubbornness - before control returns. A fall is scored: the post-fall card shows fall height, bounces, junk displaced, and a letter grade from C to SSS, with the best falls of a session collected into a rolling highlight strip on the level-select screen. This is the word-of-mouth engine in miniature: the game literally grades your funniest failure, and the share-card system of Chapter 15 turns that grade into a PNG the player can post. No fall anywhere in the game produces injury imagery, blood, or distress - MAGNO is metal, the world is junk, and every collapse ends in a bounce.

5.7 Session Zero

The first 180 seconds are specified to the second because Poki's doctrine assigns them nearly all lifetime value. Second 0-6: loading bar with the studio logo and one animated silhouette of MAGNO mid-flip - then the game starts immediately on the Scrapyard's base platform, no menu, no title screen. Second 6-20: the world teaches stick and flip with three glowing handholds and zero text; first clear of the tutorial micro-climb lands at roughly second 45 with a full celebration (confetti sparks, chime, star stamp). Second 45-180: levels 1-2 play as extended tutorials with guaranteed clears, the first cosmetic unlocks free in the locker, and the first midroll opportunity arises only after level 2's natural stop. The composite engagement signal is designed to peak at each level clear and never dip below the quit-risk line between peaks, per the shape below.

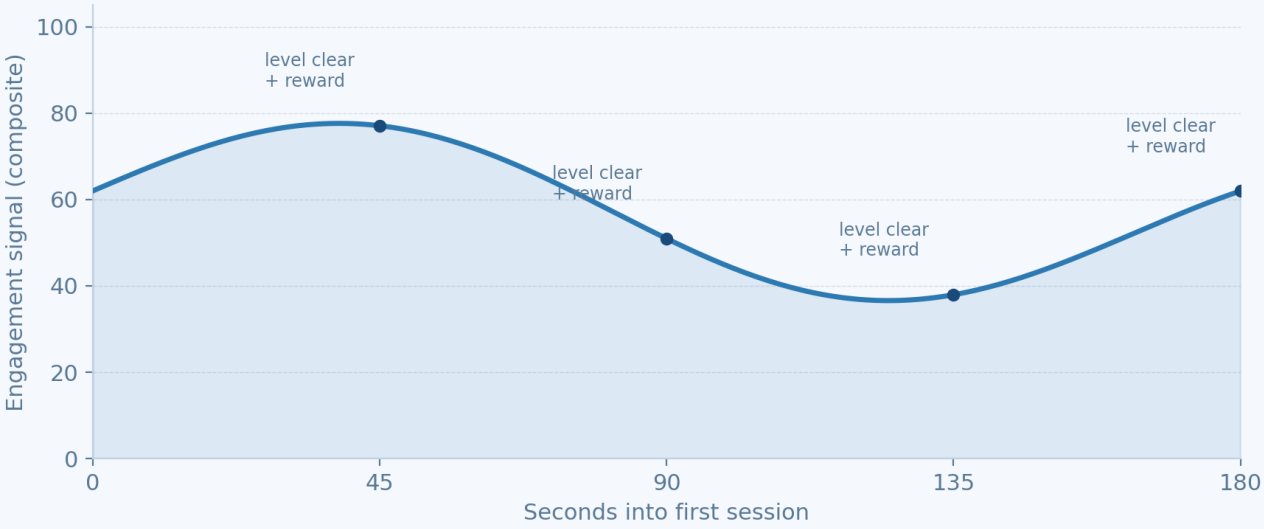


Figure 5.3 - Session-zero engagement shape: peaks at every level clear, no trough below quit risk

5.8 Content Scope

Content	Count	Notes
Levels	70 (5 biomes × 14)	30-90 s each; every 3rd-4th introduces a new prop or twist
Distinct props with unique physics	60	40 static-material, 20 dynamic/reactive
Cosmetics (MAGNO skins + field-line colors)	24 at launch	12 Scrap-purchasable, 8 reward-path, 4 seasonal
Ghost types	3	Par ghost, personal ghost, friend/regional ghost
Music stems	5 biome sets × 4 stems	Altitude-adaptive layering (Chapter 9)
Locale coverage	7 languages	en, de, es, fr, it, pt, tr (Chapter 12)

Table 5.2 - Launch content inventory (~60-75 minutes of first-run play)

6 Art Direction Domain

6.1 Style Bible

The visual language is **stylized low-poly flat-shaded 3D** with a toy-scale diorama feeling: models live between 200 and 900 triangles, every prop reads as a chunky collectible rather than a simulation, and the whole mountain composes like a shelf of oversized knick-knacks. Flat shading (one normal per face, vertex colors preferred over textures wherever a prop is a solid color) gives the game its two free superpowers: tiny file sizes and instant silhouette readability at thumbnail scale. Detail lives where gameplay lives - a fridge door has a visible handle seam because the handle is a grabbable edge - and nowhere else. Nothing in the game uses photoreal textures, normal maps, or photographic source material, which keeps the art style defensibly original and cheap to stream.

Three rules govern every asset review, and any model that fails one is reworked rather than excused. First, the **silhouette test**: a flat black render of the prop must be identifiable to a playtester in under a second, because players read shapes before colors and mobile screens before shapes. Second, the **material truth test**: the prop's magnetic class must be guessable from geometry alone - steel looks dense and riveted, weak-metal looks thin and crumpled, non-metal looks matte or soft. Third, the **budget test**: the model, compressed, must fit its slot in the per-biome asset budget of Chapter 10, or it does not ship regardless of quality.

6.2 Color and Lighting System

Color is narrative: the mountain tells its story through a temperature journey from warm ground-level junk to cool stratospheric silence, and every biome owns a dominant family with a single contrasting accent used only on gameplay-relevant edges (grabbable ferromagnetic surfaces carry a faint cyan rim tint by default, adjustable in accessibility settings). Lighting is one directional sun with a fixed per-biome angle and a warm-to-cool color story, plus a hemisphere fill; there are no dynamic shadows on mobile tier - a baked vertex AO pass at import time provides depth for near-zero cost. Distance fog is per-biome and doubles as both mood and a draw-distance curtain: rust haze in the Scrapyard, laundry-soft white in Suburbia, blue dusk fog in Metropolis, chalky glare in the Playground, and near-vacuum star-clear air in the Stratosphere.

Biome	Dominant family	Accent	Sun angle & color	Fog
Scrapyard	Rust orange H≈22°, S 55%, L 45-60%	Teal H≈188°	Low 25°, warm 3200 K	Dust haze, 15% density
Suburbia	Mint cream H≈140°, S 30%, L 75%	Coral H≈12°	High 55°, neutral 5600 K	Soft white, 8%
Metropolis	Cool slate H≈215°, S 35%, L 40-70%	Neon magenta H≈315° (signage only)	Dusk 10°, 2400 K rim	Blue, 18%
Playground	Bleached candy H≈345°/48° dual, S 40%, L 70%	Sun yellow H≈48°	Overhead 70°, flat 6500 K	Glare, 10%
Stratosphere	Indigo H≈245°, S 45%, L 30-55%	Aurora cyan H≈180°	Ambient starlight, no sun	None (star clear)

Table 6.1 - Per-biome palette contract (hue in HSL terms)

6.3 Character Design: MAGNO

MAGNO is engineered as an animation rig first and a mascot second. The body is a rounded capsule (roughly 0.6 m tall, 0.45 m diameter) whose round bottom guarantees that every fall ends in a roll or a wobble instead of a crunch - slapstick is baked into the skeleton. The head is an oversized horseshoe magnet with chrome-red tips, tilted at a permanent 8-degree confident angle; the face is a single large eye in a dark socket that handles the entire expression system (squash for determination, wide for falling, half-lid for smug). Arms are two-segment noodles with spherical hands that grab automatically during stick. Total triangle budget: 480 for body + head, 120 per arm. The silhouette is a squat bowling-pin with a U on top - identifiable at 32 pixels, which is exactly the size the thumbnail demands. Skins recolor the chrome ring, the eye socket, and add small clip-on props (antenna, sticker slap, tiny flag); no skin ever changes silhouette, preserving gameplay readability and zero-cost thumbnail consistency.

6.4 Environment and Prop Language

Every biome's prop library is authored as a family of 12-15 master props with modular variants, targeting 60 unique physics identities across launch (Table 5.2). Design language rules keep the world coherent: all junk shares a scale story (objects are 1.2-2× real-world scale so MAGNO always looks like a toy among toys), all metal carries rivets or seams, all weak-metal carries dents, and all interactive surfaces exaggerate their affordance (a grabbable handle is 1.4× normal size). The 20 dynamic props are the comedy engine: girders that see-saw, washing machines that boing, filing cabinets that shed drawers, playground swings that act as pendulums, a ferris wheel that turns on MAGNO's field. Each prop's physical parameters (mass, friction, restitution, magnetic class) live in one data table that both the artists' Blender kit and the runtime consume, so visual identity and physics identity can never drift apart.

6.5 Production Pipeline (AI-Assisted, Hand-Finished)

The pipeline is built to clear Poki's AI-production bar with documentation rather than luck. Concepting uses generative tools for exploration only - silhouette sheets, palette studies, prop ideas - and every generated

concept is redrawn or re-modeled by hand in Blender before it can enter the game; raw generator output never ships, and no asset carries watermarks or prompt artifacts. All prompts and iterations are logged in a per-asset record (prompt, tool, what was changed by hand, and why) so the creation process can be demonstrated to Poki on request, exactly as the guidelines require. Modeling is Blender with a strict export recipe: glTF 2.0, Meshopt compression, vertex colors for flat-shaded surfaces, KTX2/Basis textures only where a prop genuinely needs an albedo map (target: fewer than 25% of props). An import script validates every asset against the three style tests plus its byte budget automatically, and rejects failures at cook time rather than review time.

6.6 Thumbnail Design

The static thumbnail is the single highest-leverage 628×628 pixels in the project, so its design is specified as tightly as any level. Composition: MAGNO mid-flip in the lower-left third, field lines arcing toward a towering fridge-girder stack that fills the upper-right two-thirds, sky in the biome-1 palette; motion is implied by a 15-degree body tilt and a spark trail. Style must match the game exactly (Poki's guidance: a mismatched thumbnail hurts conversion twice - at click and at play). No text, no borders, no letterboxing, full-bleed square at 628×628 minimum with corners intact since Poki rounds them; contrast deliberately avoids the Poki playground background color (#83FFE7-adjacent hues are banned from the thumbnail palette). The animated thumbnail (required before global release) is a 3-second loop of the same composition: flip, spark, catch - captured in-engine at 60 fps from a camera path authored alongside the level 1 kit so it stays perfectly on-brand.

7 Animation Domain

7.1 Philosophy: Physics Is the Animator

In a ragdoll physics game, the simulation supplies locomotion and the animation layer supplies **intent and personality**. MAGNO's motion is a three-layer system: the physics body does the falling, the procedural animation layer does the reacting, and a small library of hand-authored poses stamps readable emotion on top. The craft goal is that a spectator watching a silent clip can tell the difference between a player's panicked flail and a confident ascent purely from secondary motion - arms trailing on fast flips, windmilling during long falls, a smug idle drum of the fingers on level clear. Every animation decision serves either readability (what is this body doing) or charm (who is this body), and anything serving neither is cut.

7.2 Squash and Stretch Contract

Squash and stretch is the style's heartbeat, and its amounts are specified numerically so they stay consistent across thousands of procedural contexts. Impact squash on attach: scale to 85% along the contact normal for 90 ms with a 140 ms elastic recovery. Release stretch on flip: 118-124% along the impulse vector, proportional to charge power, recovering over the first 200 ms of flight. Long-fall stretch: a gentle 108% vertical while in fast free-fall, sold further by the eye narrowing. Landing-from-height squash bottoms at 72% with a dust ring scaled to impact speed. The magnet head participates inversely (stretch when the body squashes) so total volume reads as preserved - the classic principle applied at rig level rather than keyframe level, since almost no

animation in this game is keyframed.

7.3 The Ragdoll Blend System

MAGNO exists on a continuous spectrum between four body states, blended by weights rather than switched. **Active** (full motor control, spring forces hold the capsule upright and arms track targets), **Scramble** (a hit of chaos: control authority dips to 40% for 400 ms after hard contacts - the wobble that reads as being rattled), **Limp** (all motor forces off, pure ragdoll, engaged for falls and big impacts), and **Recover** (a 650 ms spring-powered get-up sequence that is procedurally initiated from whatever pose the body settled in, using torque-to-upright plus a two-beat arm push-off, ending in a 120 ms settle bounce). The get-up is the most important animation in the game - it plays after every fail - so it is authored as a parameterized system (spring constants ramp, head tilt correction, eye re-focus) rather than a canned clip, ensuring it never repeats identically and never clips through geometry because it is computed from the actual resting pose.

7.4 IK Arms and Secondary Motion

Arms are two-bone IK chains driven by target points that lag the body's motion by 80 ms, which is the entire secret of weight: fast direction changes make the arms trail, stops make them overshoot, and the character instantly feels massy. Hand targets have three behaviors - grab-seek (during stick approach, hands reach toward the surface within a 30° cone), free-float (a damped noise orbit around the shoulder line during flight, amplitude scaled by airspeed), and brace (pre-impact anticipation pose 150 ms before predicted landing). The head runs its own smaller lag chain (30 ms) with magnetic orientation bias toward the current target surface, so the eye naturally leads the player's intent - a subtle, constant readability aid.

7.5 UI Motion System

Token	Duration / curve	Used for
micro	120 ms, standard ease-out	Button presses, star stamps, ticker ticks
short	220 ms, cubic ease-out-back	Panels in/out, level cards, locker grid
medium	380 ms, quintic ease-in-out	Screen transitions, biome banners
hero	600 ms, custom overshoot 1.12	Level-clear banner, fall grade card, unlocks
loop	1.8 s sine, ±4 px	Idle ambient motion (beacon pulse, flags)

Table 7.1 - UI motion tokens (all tweens use these, nothing hand-timed)

7.6 Frame Budget

System	Budget	Notes
Ragdoll solve + IK chains	0.8 ms	6 joints, closed-form IK; no full-body solver
Blend state machine	0.2 ms	Weight math only; no clip sampling in play
Secondary motion (lag chains)	0.15 ms	Damped springs, allocation-free
UI tweens	0.3 ms	Single scheduler, transform-only
Total animation cost	≤ 1.5 ms	Leaves 14+ ms of the 16.6 ms frame for physics + render

Table 7.2 - Animation CPU budget at 60 fps on the mid-tier reference device

8 VFX Domain

8.1 The Magnetic Identity

All signature effects serve one visual thesis: **magnetism made visible**. The field-line system renders animated dipole curves flowing around MAGNO's head - blue-white in attract mode, orange-gold in repel - drawn as a shader ribbon strip whose intensity scales with charge. On flip, the lines snap-invert with a 120 ms radial pulse; on snap, they stretch toward the target surface and bend as the pull engages. Because polarity color is consistent everywhere (HUD charge meter, field lines, snap-target highlight, head glow), the player always knows their own state in peripheral vision, and spectators can read a clip with the sound off. The palette is deliberately two-note: cyan-family and amber-family on the biome-neutral page of the world, so effects never fight the environment art.

8.2 Effect Inventory

Effect	Description	Peak particles	Trigger
Field lines	Dipole ribbon shader, animated flow, polarity-colored	24 segments	Always on, intensity by charge
Attach spark burst	Contact-point spark fan + one ring wave	40	Every stick
Repel shock ring	Inverted ring + 6 directional wisps	22	Every flip
Snap trail	Ribbon from body to target, tapering	36	During pull
Debris avalanche	Loose junk physics wake with dust puffs	dust 90 / debris = physics bodies	Falls through prop zones
Slow-mo vignette + speed lines	Radial streak overlay at 8% opacity	screen-space	Falls > 12 m
Level-clear constellation	Stars stamp with confetti sparks + beacon flare	80	Clear

Effect	Description	Peak particles	Trigger
Wind gusts	Bent-line streaks, biome-colored	18	Metropolis+ biomes

Table 8.1 - Signature effect inventory with budgets (particle counts are max concurrent)

8.3 The Slow-Motion Fall System

Big falls are the game's signature shareable moment, so they are produced like a shot in a film rather than left to chance. Trigger: fall height above 12 meters or impact speed above 14 m/s. On trigger, time scale ramps to 0.35 over 150 ms, the camera dollies back and up to frame the arc, a radial speed-line overlay fades in at 8% opacity, the falling whistle drops in pitch with the timescale (audio and simulation share one clock), and field lines trail the body in limp mode. On impact, a single frame of white flash at 5% opacity, dust ring, time scale recovers over 450 ms, and the fall-grade card slides in. The system has a per-session budget (maximum 4 slow-mo falls per 3 minutes, prioritizing the biggest) so the effect never becomes noise, and it is fully disabled by the reduced-motion accessibility setting, which substitutes a fixed wide camera with no time distortion.

8.4 Screen Shake Grammar

Shake is a language with a strict grammar, not a volume knob. Micro-shake (amplitude 0.10 of camera height, 80 ms): stick attach on weak metal, prop collisions nearby. Small (0.22, 120 ms): charged flip release, dynamic prop topple. Big (0.38, 180 ms): high-speed impact, avalanche events. Shake is positional noise in the 8-12 Hz band with rotational component capped at 1.2 degrees, exponential decay, and a hard rule that it never triggers during snap-in-progress or any input-critical window, because ruining aim is the cheapest way to make a physics game feel unfair. The reduced-motion setting reduces all amplitudes by 70% and removes rotational shake entirely.

8.5 Post-Processing and Mobile Fallback

The desktop post stack is deliberately short: a soft bloom (threshold 0.85, strength 0.4 - used only on beacons, the core, and field lines), a 12% vignette, and a per-biome grading LUT authored in Blender's film color tools and baked to a 256×16 texture. The mobile tier drops bloom to half-resolution or disables it (a quality flag decides at device-detection time), keeps the LUT (one texture fetch, effectively free), and replaces the vignette with a pre-baked corner darkening in the biome's gradient fog. No depth-of-field, no motion blur, no chromatic aberration: each was tested and cut - the first two cost frames on low-end devices, and aberration reads as rendering error at small scale rather than style.

8.6 Particle Budget by Device Tier

Tier	Examples	Total particle ceiling	Fallback
High (desktop, recent phones)	RTX/Mali-G710 class	900	None
Mid (reference device)	Snapdragon 695, A13	500	Dust systems halve; trail tapers sooner
Low	Snapdragon 460, 2019 Androids	260	Field lines reduce to 12 segments; avalanche dust off

Table 8.2 - Concurrent particle ceilings enforced by the VFX director at runtime

9 Audio Domain

9.1 Philosophy and the Anti-Slop Moat

The audio direction exists to make MAGNETOPE sonically undeniable as a handcrafted product in a market flooded with stock-loop games. Every surface has a material-coded voice; every system state has a continuous sonic signature; the music is a set of adaptive stems that respond to altitude and momentum; and the total shipped audio footprint is under 350 KB because nearly everything is synthesized at runtime from oscillators, noise, and filters rather than shipped as recordings. This is not an austerity measure - it is the sound of the game's identity: a world built from junk should sound assembled from raw elements, precisely tuned. It also happens to be the single strongest answer to "this looks like every AI game": no other submission will sound like this one.

9.2 SFX Inventory

Event	Recipe sketch	Feel target
Attach: fridge/washer body	180 Hz sine thunk + 900 Hz noise burst (12 ms) + body resonance tail 140 ms	Deep, domestic, satisfying
Attach: girder/beam	420 Hz metallic ring, two detuned partials, 260 ms decay	Industrial chime
Attach: sheet/weak metal	Crumple: filtered noise sweep 3 kHz→800 Hz	Nervous, thin
Flip release (tap → full)	Air whoosh: band-passed noise, center 500→1800 Hz by power; sub thump at full	Breath in, launch out
Snap engage	Rising zap: sawtooth 220→880 Hz over 90 ms + hum swell	Magnetic pull made audible
Snap catch	Same attach recipe as target material + confirmation tick	Relief
Fall whistle	Sine glide 1200→300 Hz, pitch mapped to fall height, vibrato in slow-mo	Cartoon inevitability

Event	Recipe sketch	Feel target
Impact (per material)	Material attach recipe pitched down + layered crunch noise	Weight and comedy
Get-up spring	Boing: damped sine bounce 90→160→130 Hz	Dignity restored
Level clear fanfare	3-note arpeggio (beacon motif) + confetti sparks noise	Genuine celebration, 1.2 s
UI micro	Tick: 1 ms filtered click at 2 kHz	Crisp, never cute

Table 9.1 - Core SFX inventory (all synthesized at runtime; layer recipes in the audio DSL)

9.3 The Polarity Hum System

The game's most distinctive continuous sound is the magnet itself. In attract mode, a low hum of 55 Hz + 110 Hz partials swells with proximity to ferromagnetic mass - the player literally hears how close the next hold is, which doubles as a soft assist and a tension device while dangling. In repel mode the hum crossfades to a detuned sawtooth shimmer whose brightness tracks charge level. When charge is full, a barely-perceptible 16 Hz tremolo makes the held flip feel physically tense. The hum is mixed at -30 LUFS-relative under normal play so it colors rather than narrates, and the accessibility options offer a visual charge meter replacement for hearing-impaired players.

9.4 Adaptive Music Architecture

Each biome ships four stems - **ground** (bass/percussion loop), **layer 1** (melodic motif), **layer 2** (harmonic pad), and **event** (stingers: clear, big-fall, unlock) - authored in one key per biome so any layer combination is consonant. The music director follows a deterministic rule set: ground always plays during a run; layer 1 fades in above 40% of level height; layer 2 engages during snap-chain streaks (three surfaces in under six seconds); during slow-mo, all stems low-pass and stretch with the timescale; on clear, stingers resolve to the biome's tonic. Transitions are 400 ms equal-power crossfades, never instant cuts. The whole score is delivered as Opus-encoded stems (48 kHz mono, ~40 KB per stem), decoded at load - approximately 800 KB across all five biomes, streamed in the background with the biome chunks rather than in the initial payload.

9.5 Ambience and Wind

Ambience is one filtered-noise wind voice whose character evolves with altitude: Scrapyard ground level is barely audible warm air with distant clanks (randomized metal creak events triggered by physics contacts far below), Suburbia adds synthetic birdsong blips, Metropolis layers a hum of neon buzz, and Stratosphere strips everything to a thin whistle and eventual silence - the quiet itself becomes the biome's identity. Gusts are shared with the gameplay wind system, so audio wind and physical wind are one parameter, and the player can hear a gust coming before it pushes them, which is both fair play and mood.

9.6 Implementation Architecture and Mix

```
// Web Audio graph (single AudioContext, unlocked on first input)
ctx = new AudioContext(); // after PokiSDK.gameplayStart()
master -> compressor -> destination // -14 LUFS integrated target
|- sfxBus (-12 dB) : one-shot voices, material-coded recipes
|- musicBus (-18 dB) : 4 biome stem players, crossfade 400 ms
|- ambBus (-24 dB) : wind noise + biophony events
'- uiBus (-20 dB) : ticks, stingers
duck(musicBus, -6 dB, 120 ms) // during commercialBreak / rewardedBreak

// Material voice example: fridge attach
voice = synth('sine', 180Hz, decay 140ms)
+ noiseBurst(900Hz band, 12ms)
+ convolve(impulseSmallRoom) // generated 64-tap IR, zero download
play(voice, sfxBus, gainBy(impactSpeed))
```

Mix standards are fixed before content production begins so every asset is authored against them: master at -14 LUFS integrated with a true peak ceiling of -1.5 dBTP; SFX bus at -12 LUFS-relative as the loudest element; music at -18 ducking to -24 during any ad playback (a Poki-quality requirement so midrolls never fight the game); UI at -20; ambience at -24. All synthesis voices are pooled and allocation-free after load. The game must remain perfectly playable muted - audio is an enhancement layer, never an information gate - and the QA chapter includes a full silent-mode pass for exactly that reason.

10 Technology & Engineering Domain

10.1 Stack Decision

The stack decision is made against three hard constraints from the research: a ≤ 5 MB initial download, sub-10-second playable load on mid-tier mobile hardware, and a curated platform whose engine guide explicitly warns that some engines merely "support" the web rather than being optimized for it. Poki's own comparison data puts an empty Unity WebGL build at roughly 11 MB before optimization, and independent 2026 benchmarks measure Three.js-class experiences loading in 2-6 seconds where Unity WebGL commonly takes 8-30 seconds before interaction. Godot's web export remains a moving target and Defold - Poki's official partner engine at 1.03 MB for an empty project - is a strong contender, but its physics story (Box2D 2D / Bullet 3D via extensions) does not match a design whose soul is a momentum-based 3D dynamic-body controller with continuous collision on the player.

Option	Empty build (Poki data)	Physics fit for MAGNETOPE	Verdict
Three.js + Rapier WASM (TypeScript)	~0.4-0.7 MB core, tree-shaken	Rapier: dynamic bodies, joints, CCD - exactly the design	Chosen: full control, smallest ceiling, web-native
Defold	1.03 MB	Box2D/Bullet via extensions; 3D dynamic controller is a fight	Strong runner-up; revisit if web export of Three path fails
Unity WebGL	~11 MB unoptimized	PhysX excellent	Rejected: load size and time vs Poki's ceiling
Godot 4 web export	2-5 MB optimized	Good 3D physics	Rejected: web not the priority path; export instability
Babylon.js	comparable to Three.js	Good	Equivalent candidate; Three chosen for ecosystem depth

Table 10.1 - Engine decision matrix (Poki engine-guide data + 2026 web benchmarks)

10.2 Architecture

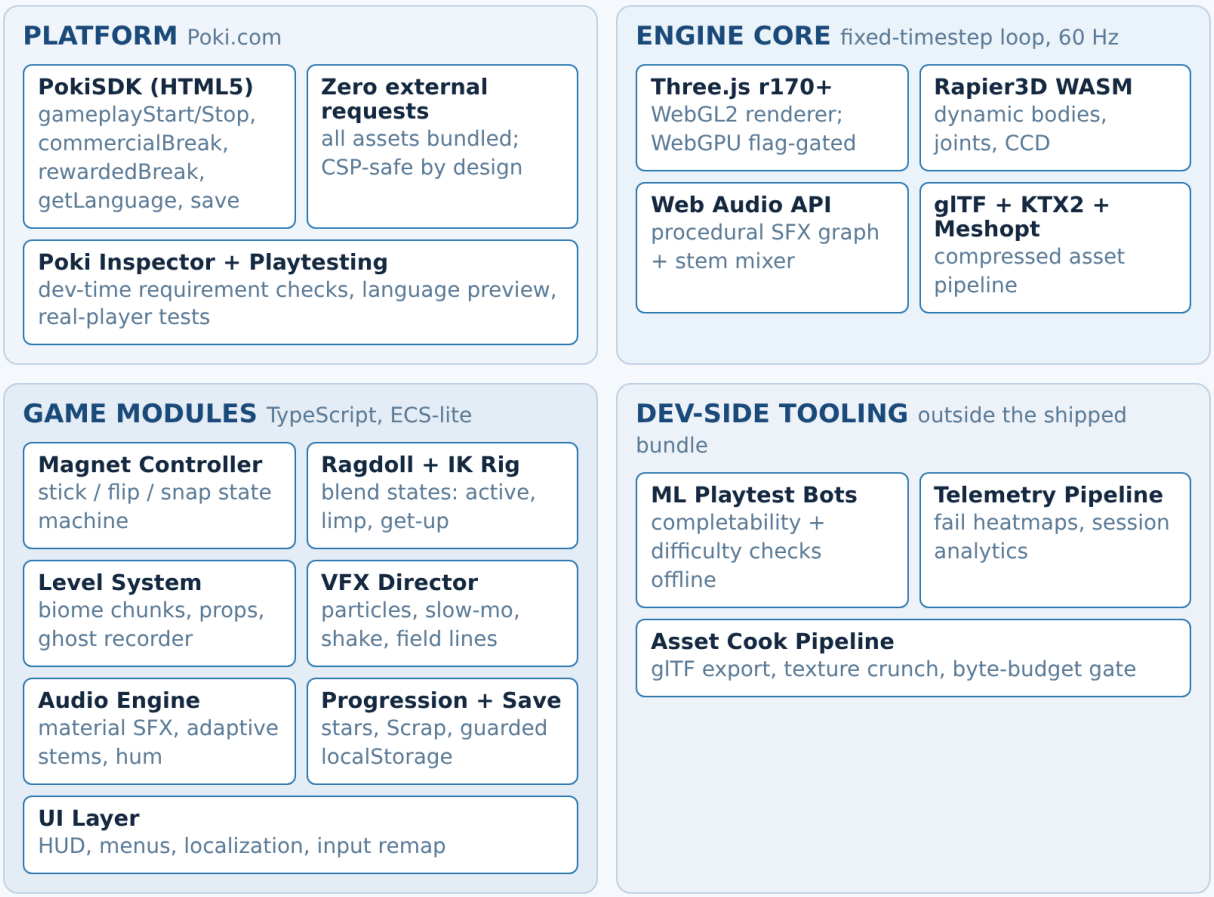


Figure 10.1 - System architecture: platform, engine core, game modules, and dev-side tooling (outside the bundle)

The game is structured as an ECS-lite: component stores (plain typed arrays and maps), systems that run in a fixed order, and zero inheritance hierarchies in the hot path. The game loop is a fixed-timestep design - simulation at exactly 60 Hz with an accumulator, rendering decoupled with interpolation of visual transforms - because deterministic physics is a correctness requirement, not a nicety: the ML playtest bots of Chapter 11, the speedrun ghosts, and the fail-replay capture all depend on the same simulation stepping identically given the same inputs. The render side is one WebGL2 renderer instance with a scene graph per biome; a WebGPU path sits behind a runtime flag for future investigation but WebGL2 is the shipping baseline since WebGPU is only now reaching default-on status across Chrome, Edge, and Firefox and is still rolling out in Safari.

10.3 Build Budget and Progressive Loading

The byte budget is treated as a design constraint with the same authority as the difficulty curve. The initial payload - everything needed to reach gameplay and clear the first three levels - totals 3.85 MB against the 4.0 MB internal target and the 5.0 MB Poki hard limit. Biome chunks (meshes, level data, music stems, locale files beyond EN) stream in the background after session start, exactly the progressive-loading pattern Poki calls "one of the most important techniques for success on the web"; the player is already playing while biome 2 arrives. Total shipped size across all chunks is engineered to stay at or under 7 MB against the 8 MB ceiling. The budget is enforced mechanically: the asset cook pipeline and the CI build both fail if any chunk exceeds its allocation, so size can only regress through a deliberate decision, never through drift.

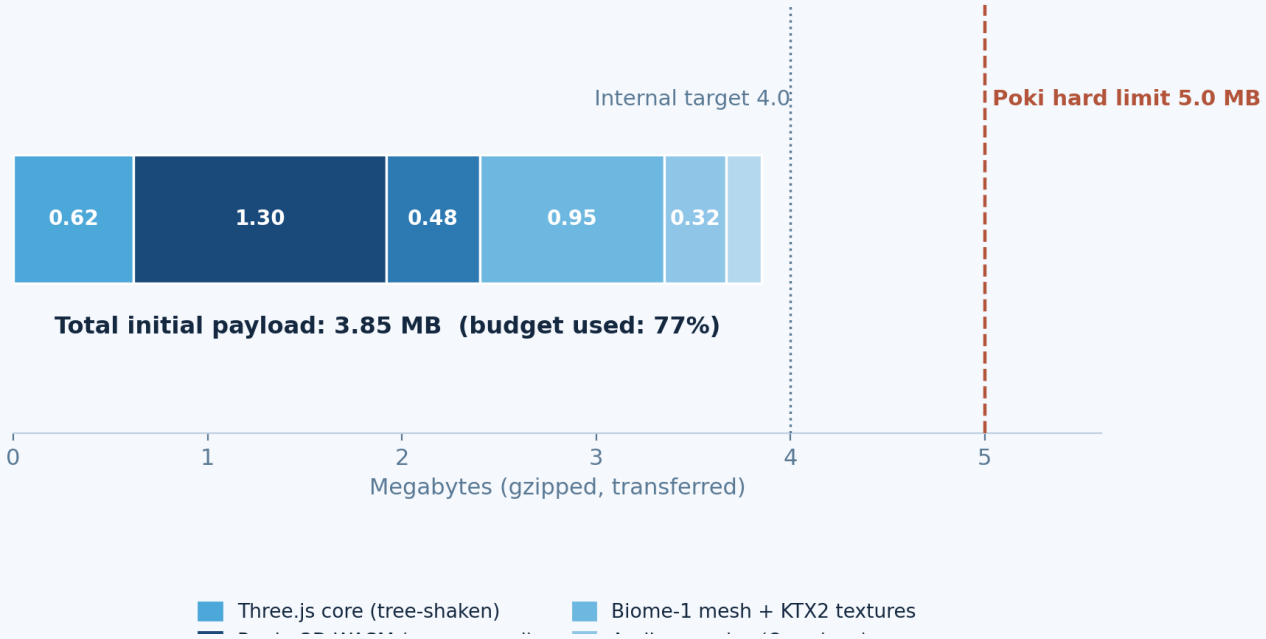


Figure 10.2 - Initial load budget breakdown vs the 4 MB internal target and Poki's 5 MB hard limit

10.4 Performance Targets

Tier	Reference hardware	Frame target	Bounded by
High	Desktop w/ discrete GPU; iPhone 15 class	60 fps locked	Physics step ≤ 3.5 ms; render ≤ 8 ms
Mid (reference)	Snapdragon 695; iPhone 12; A13	60 fps typical, 45 floor	Draw calls < 120; triangles < 350 k visible
Low	Snapdragon 460; 2019 Android; old Chromebooks	30+ fps floor	Dynamic resolution 0.55-1.0×; particles 260 cap

Table 10.2 - Performance contract per device tier (enforced by the QA protocol)

The toolchain for hitting those numbers is conventional and boring by design: instanced meshes for repeated junk props, one shared flat-shaded material per biome (plus the field-line ribbon material), frustum culling with a per-biome static cell index, dynamic-resolution scaling driven by a 30-frame rolling average of frame time, and a hard cap of 260 concurrent physics bodies with debris promoted from visual-only to physical only inside the camera frustum. Texture memory stays under 96 MB with KTX2-compressed assets. All numbers are measured nightly on the device matrix by the telemetry harness described in Chapter 11 rather than trusted from the desktop dev machine.

10.5 PokiSDK Integration

Game moment	SDK call	Rule from Poki docs
First input of a session	gameplayStart()	Must fire on first input, never on load
Pause menu opens / level end / fall card	gameplayStop()	Fires on every gameplay interruption; never twice in succession
Closing pause menu back into gameplay	commercialBreak()	Only on the pause-to-gameplay transition; never into a level-select screen
Level clear card: reward offer	rewardedBreak() (via button)	Always optional; never blocks core loop; natural option sits primary
Session start (before first input)	PokiSDK.getLanguage()	Drives locale selection with EN fallback
Progress writes	PokiSDK.save() + localStorage try/catch	Incognito-safe: in-memory fallback, game stays fully playable

Table 10.3 - SDK event map (correct implementation is a QA gate, misfires are lost revenue)

```
// Event broker - the ONLY module allowed to touch PokiSDK
class SdkBus {
state: 'menu' | 'playing' | 'paused' | 'ad' = 'menu';
start() { if (this.state !== 'playing') { PokiSDK.gameplayStart(); this.state='playing'; } }
stop() { if (this.state === 'playing') { PokiSDK.gameplayStop(); this.state='paused'; } }
resume() { if (this.state === 'paused') { PokiSDK.commercialBreak().then(() =>
this.state='playing'); } }
rewarded(): Promise { return PokiSDK.rewardedBreak(); } // music ducked -24 dB meanwhile
}
// CI trace test: 10k scripted inputs -> assert zero double-fires,
// zero events during ad windows, start fires on input #1 only.
```

10.6 Scaling, Storage, and CSP Compliance

Rendering scales to Poki's required canonical sizes (640×360, 836×470, 1031×580) by letterboxing the fixed 16:9 logical stage inside the available viewport, choosing the nearest canonical resolution at or below the device's CSS pixels, and covering the full canvas so there are no visible bars on any device. Storage writes wrap every localStorage access in try/catch: on quota errors or incognito restrictions, the game swaps to an in-memory store, continues fully playable, and surfaces a single soft toast that progress will not persist - exactly the pattern Poki's requirements demand. The build performs zero external network requests at runtime: fonts are a bundled subset, all assets ship in the bundle, and the only optional network surface (ghost exchange) uses Poki's own AUDS platform API, which is first-party and pre-approved rather than an external service requiring CSP exception review.

11 Machine Learning Domain

11.1 Policy: Intelligence Off the Wire

The ML policy is stated as an architecture constraint before it is a capability list: **no model ships in the game bundle**. A WebGPU/WASM inference runtime alone would consume 2-8 MB of the 8 MB total budget and add first-load latency, which would trade the game's two hardest-won platform advantages (size and speed) for features the player cannot see. Every ML component therefore runs on the development machine, against the deterministic fixed-timestep simulation, and its outputs ship as data: difficulty curves, level flags, telemetry-derived thresholds, and balance patches. The player-facing game stays a hand-tuned 3.85 MB artifact; the intelligence lives in the pipeline that produced it.

11.2 Playtest Bots

The flagship ML tool is a reinforcement-learning playtester. A PPO agent trains in a headless build of the real game - same Rapier physics, same magnet controller, rendered nothing - on a curriculum of individual level segments, receiving reward for altitude gained, level completion, and (weighted negatively) fall count and wall-clock time. Training uses domain randomization over input jitter and reaction latency so the resulting policy approximates a sloppy human rather than a frame-perfect robot. Each trained policy then plays every level 1,000 times overnight, and the harness reports per-level: completability (any policy solves it), median attempts-to-clear, fall-rate distribution, and the exact 3D positions where falls cluster. Levels that no policy completes are broken by construction; levels whose fall rate drifts outside the curve contract of Figure 5.2 are flagged into the designer's review queue. This closes the loop that makes the difficulty chapter enforceable before a human ever playtests.

11.3 Telemetry and Difficulty Analytics

The shipped game emits a compact anonymous event stream - level entered, stick/flip/snap counts with positions, fall events with height and landing cell, clear times, retry latency, session length - batched in memory and flushed to the Poki dashboard's event system on natural stops (never mid-run, never during ads, and only via first-party SDK channels so the CSP story stays clean). Development-side tooling clusters fail positions per level (k-means over fall coordinates, k selected by silhouette score), renders them as heatmaps onto the level's top-down layout, and mines retry sequences to find the specific obstacle where players quit rather than retry. Weekly, the heatmap review answers three questions per level: where do players fail (is it the designed twist), is the fail readable (do they understand why), and does the fail rate sit inside the contract band. The same pipeline, pointed at soft-release data, becomes the live difficulty-tuning engine described in Chapter 15.

11.4 Generative Assist and Poki's Ownership Bar

Generative models accelerate three production surfaces under the strict Poki AI policy established in Chapter 2.4 and the asset pipeline of Chapter 6.5: concept exploration (silhouette sheets and palette studies, redrawn by hand before any model enters the game), boilerplate code (tests, build scripts, GLSL first drafts - always reviewed, profiled, and owned by the developer), and drafting (this document's structure, store copy, and locale

string first passes, rewritten by the developer before shipping). Every use is logged in the asset ledger with prompt, tool, output, and the human refinement that followed, so the creation process can be demonstrated to Poki on request and IP ownership remains unambiguous. Nothing generated is ever shipped unreviewed; no prompt text or watermark can appear anywhere in the build; and every level - regardless of how it was prototyped - is hand-balanced and bot-verified before release.

11.5 Data Ethics

- Zero PII: no accounts, no emails, no sign-ins, no chat, no free-text input anywhere in the game.
- All telemetry is event-level gameplay data with no device identifiers beyond what Poki's own SDK provides.
- GDPR-friendly by architecture: nothing to export because nothing identifying is collected.
- Bots and analytics run on data the developer already owns; no third-party analytics services are integrated (Google products are explicitly non-approvable on Poki).

12 UX & UI Domain

12.1 Onboarding Contract

Poki's easy-access guidance - skip the menu, start safe, teach visually, introduce gradually - is implemented as a second-by-second contract rather than a philosophy statement. The game boots directly into the Scrapyard base platform with zero interstitial screens; the loading bar itself (logo, progress, one animated MAGNO silhouette) is the only pre-play moment that exists. The first 20 seconds contain three glowing handholds, one animated ghost-hand demonstrating stick-and-flip (no text), and a first micro-clear at about second 45 with full celebration. The pause menu and level-select are reachable but never forced; the HUD fades in elements only as they become relevant (altitude at first vertical gain, charge meter at first hold-to-charge at level 3, star tracker after the first clear). New mechanics continue to arrive on the drip schedule of Table 5.1 - one new idea per 3-4 levels - for the entire first biome.

12.2 HUD and Screens

The play HUD is deliberately near-invisible: altitude (a slim vertical rail on the left with a summit marker), charge meter (an arc around MAGNO, polarity-colored), and the level timer that appears only after first clear (speedrun chasing is opt-in signaling, not pressure). No coins counter during play, no combo popups, no minimap - Scrap payouts and grades live on the level cards where they belong. The complete screen inventory is six surfaces: loading, play HUD, pause (resume, restart, sound, language, quit-to-map), level-select map (the mountain itself, zoomable, with fall-grade highlights), level-clear card (stars, time vs par, Scrap, next-level and reward offers), and the cosmetic locker. Every surface is fully navigable by keyboard, mouse, and touch with the same focus order, satisfying Poki's input guidance (WASD/arrows through menus, space/enter as confirm).

12.3 Localization

The game is engineered text-light by design - the tutorial is gestural, signage is environmental, and the HUD is iconic - which reduces the localization surface to roughly 120 strings. All text lives in a single strings.json per locale, loaded by chunk with PokiSDK.getLanguage() selecting the initial locale and English as fallback; layouts are tested against the longest natural translations (German and Portuguese) with elastic text fields rather than fixed boxes. Launch locales follow Poki's recommended priority: English, French, Italian, German, Spanish, plus Turkish (the increasingly popular locale Poki explicitly calls out), plus Brazilian Portuguese - seven locales covering the platform's broad EFIGS+TR+PT base. CJK locales are deferred deliberately: they are costlier to translate, and the game's visual-first design means non-localized players can still play everything without friction, which the Poki localization guide itself acknowledges as the low-risk posture.

12.4 Accessibility

- Mouse-only mode: full play via cursor-relative tilt and click-flip, per Poki's alternative-controls guidance.
- Touch options: left/right split layout or single-thumb tap zones, both remappable; drag-steering sensitivity slider.
- Auto-align assist on snap (Chapter 4.5) defaults on; expert players disable it for a 5% Scrap-earn bonus.
- Colorblind safety: polarity states are distinguished by hue AND shape (field lines converge vs diverge); palette verified for deuteranopia and protanopia.
- Reduced motion: slow-mo replaced with fixed wide camera, shake amplitudes cut 70% and rotation removed, per Chapter 8.
- Subtitles-adjacent cueing: all critical audio cues (full charge, snap window) have visual twins in the HUD.
- No timed text anywhere; no QTEs; every retry path is one input.

13 Monetization Domain

13.1 Model Recap and Placement Map

All monetization flows through the Poki SDK: the game signals natural stopping moments and player-chosen reward opportunities, and Poki's system decides when an ad actually serves. The design consequence is that **signaling more natural stops honestly is the entire ad strategy** - the game never decides to show an ad, it only decides where an ad would not be unwelcome. Every placement below obeys the documented rules: commercialBreak fires only on the pause-to-gameplay transition, rewardedBreak only on explicit player action, and no SDK event ever fires during another ad. Portrait orientation is supported throughout, which makes the game eligible for Gamebar Display ads - mobile title-bar revenue that requires no work and no code beyond the orientation support itself.

Moment	Call	Why it is natural	Guardrails
Level clear → next level	commercialBreak (via resume path)	Player just finished a goal; hands are off the controls	Skippable state, never mid-animation
Retry after 3+ consecutive falls	commercialBreak (pause path)	Frustration peak is a documented natural stop	Capped by Poki; never during fall physics
Fall-grade card: Second Wind offer	rewardedBreak	Player chooses: watch to resume at last checkpoint with height insurance, or retry from base	Always optional; retry button stays primary
Cosmetic locker: direct unlock	rewardedBreak	Watch-to-unlock alternative to spending Scrap	Standard video icon; reward celebration on completion
Ghost locker: friend/regional ghost data	rewardedBreak	Optional competitive depth	Never gates core play

Table 13.1 - Ad placement map (frequency is Poki-side; the game only signals opportunities)

13.2 The Scrap Economy

The economy follows Poki's monetization-tips guidance exactly: one understandable currency, no secondary currencies, clear rewards, and rewarded moments designed in from the start. Scrap earns at 10-50 per level (first clear) and 2-8 per replay by performance, and the 24 launch cosmetics are priced 150-400 so a casual player unlocks roughly one skin per biome of progress while a dedicated player can chase the full locker - ownership as investment, exactly the customization pattern Poki's own examples favor. Second Wind (the rewarded fall insurance) is deliberately not purchasable with Scrap: it is the watch-to-earn bridge that keeps the rewarded split healthy, and its offer card appears only on the fall-grade screen where the player has just experienced the exact pain it solves. The non-rewarded path (retry from base) always sits in the primary button position, per the platform's explicit best practice.

13.3 Revenue Scenarios

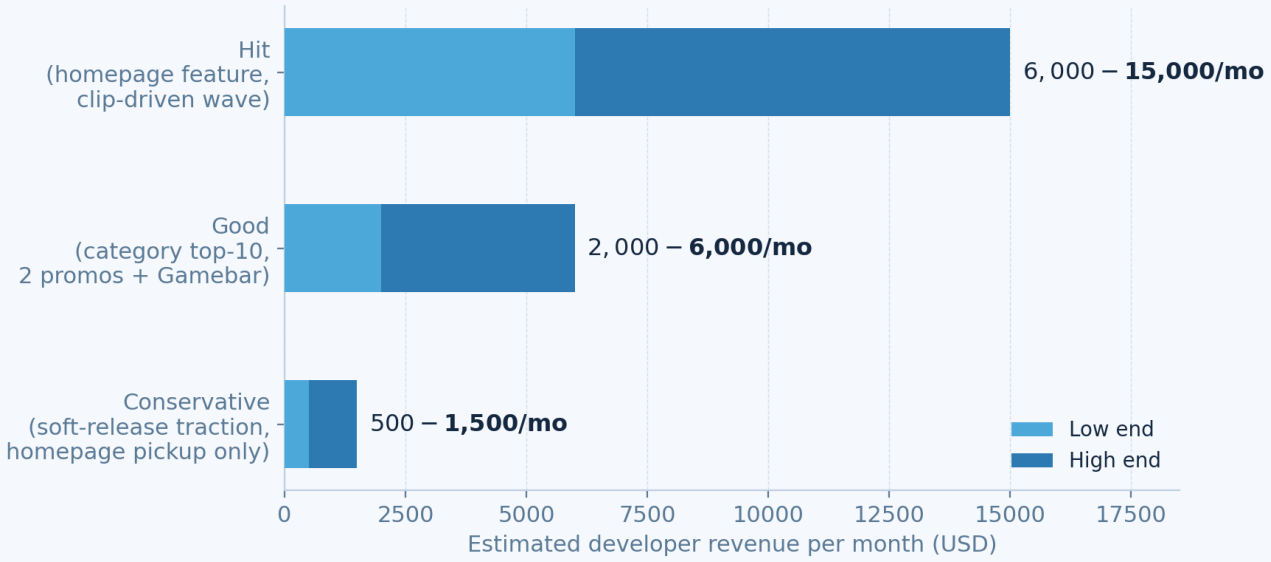


Figure 13.1 - Monthly revenue scenarios with assumptions from Table 13.2

Scenario	Entry conditions	Assumptions
Conservative: $500-1,500/mo	Passes final review; soft release steady; category pages only	12k-25k DAU, 1.0-1.2 ads/DAU, mid-tier eCPMs; matches the documented first-web-game band
Good: $2,000-6,000/mo	Top-10 in 2+ categories; two promotion cycles; Gamebar live	45k-90k DAU, 1.2-1.5 ads/DAU, rewarded split 35%
Hit: $6,000-15,000/mo	Homepage feature; clip-driven wave; ghost culture compounding	150k+ DAU, sustained sessions, seasonal update retention

Table 13.2 - Scenario assumptions (revenue-share model, Poki-driven traffic)

The funnel that produces those numbers is engineered end to end: the thumbnail earns the click (CTR), the 3.85 MB load and instant-boot flow earn the play (C2P - which Poki documents as mostly a technical story of load time, size, and responsiveness), the three-minute loop and fall comedy earn the session (engagement), and the natural-stop discipline plus the Second Wind bridge earn the ad depth (≥ 1 ad per DAU at soft release, the platform's stated healthy target). Every chapter of this document feeds one stage of that funnel, which is the point of building to the platform's playbook rather than against it.

14 QA Domain

14.1 Poki Checklist Mapping

Poki QA checks every game against the Requirements page before soft release, and passes come with itemized reports until the game is through. The strategy is to front-load that checklist into CI and the device matrix so the official QA pass finds nothing to report. The mapping below is the master test plan - every row is an automated test, a scripted device-matrix run, or a manual protocol with a named owner and a fixed cadence; the appendix repeats it as a one-page sign-off matrix for the pre-submission gate.

Poki requirement	MAGNETOPE implementation	Verification method
Desktop + mobile + tablet; full-screen coverage	16:9 logical stage, letterbox to canonical sizes, tablet forces touch UI	Device matrix pass per build (Table 14.2)
Incognito support	Storage try/catch with memory fallback + toast	Automated: launch in incognito profile, complete 3 levels, verify no errors
No external requests	Bundled font/assets; no runtime fetch except biome chunks from Poki CDN	CI: CSP-simulated sandbox run; network log must be empty
No splash/outgoing links/foreign ads	Logo on loading bar only; zero outbound anchors	Automated DOM/event audit per build
SDK event correctness	SdkBus broker (10.3)	CI trace test: 10k scripted inputs, assert event invariants
Clean build	Debug stripped via build modes; dev tools behind URL flag removed in prod	Diff audit of prod bundle symbols
Save system + notice	PokiSDK.save + localStorage fallback	Scripted save/restore across reload, quota-denied path
Size and load speed	3.85 MB initial, chunked rest	CI measures gzipped payload and cold-load time; fails over budget
Static + animated thumbnails	628×628 full-bleed + 3 s 60 fps loop, both in-engine captures	Spec sheet check + Poki Inspector preview
Keyboard pause (ESC/space)	Pause system wired to SdkBus	Automated input fuzz on all screens
Skippable cutscenes; visual tutorials	Ending skip from second 0; gesture tutorials	Manual protocol + first-session recordings

Table 14.1 - Requirement → implementation → verification

14.2 Device Matrix

Tier	Devices	Browsers
Desktop high / low	Dev desktop; 2015 dual-core laptop	Chrome, Firefox, Edge (latest)
iOS mid	iPhone 12	Safari (current + previous major)
Android mid (reference)	Snapdragon 695 class	Chrome mobile
Android low	Snapdragon 460 / 3 GB RAM	Chrome mobile
Tablet	iPad 9th gen; mid Android tablet	Safari, Chrome
Hostile profiles	Incognito; ad-blocker enabled; airplane after load	Chrome incognito, Firefox + uBO

Table 14.2 - Minimum device matrix per release candidate (each cell = 30-min scripted session + 30-min manual)

14.3 Protocols

- Performance protocol: nightly automated matrix run capturing frame-time histograms, physics-step p99, memory after 30-minute soak (leak detection), and payload size; regressions fail the build.
- Event-trace protocol: every CI build replays 10,000 scripted inputs through the real input layer and asserts the SDK invariants (start-on-first-input, no double-fires, no events during ad windows).
- Silent-mode protocol: full playthrough with audio muted - the game must remain fully completable with zero audio-gated information (Chapter 9).
- Playtest protocol: before submission, 5 external playtesters (friends, family, then Poki Playtesting's real players) with session recordings; review focuses on the first 90 seconds and the first fall.
- Bug triage: S0 blocks release (crash, broken level, SDK misfire); S1 fixes within 48 h (progression blocker, perf below floor on reference tier); S2 next content update; S3 backlog.

15 Launch & Growth Domain

15.1 The Release Pipeline

Stage	Duration	Developer actions
Submission + playtesting	days-weeks	Submit build + thumbnail; run Poki Playtesting (player fit); iterate from recordings
Web fit test	days-weeks	Fix technical findings; re-upload clean build
Final review	1-2 weeks	Team reviews results + content vs quality guidelines; prepare the clone-defense matrix of Table 3.2 and AI process ledger
Agreement	variable	Review web-exclusive vs non-exclusive terms; 5-year default exclusive rev-share recommended for this game
QA	iterative	Zero-findings target via Chapter 14 front-loading; resolve reports within 48 h

Stage	Duration	Developer actions
Soft release	2-3 weeks	Monitor daily watchlist (15.2); ship difficulty and UX fixes; A/B thumbnails; add animated thumbnail
Global release	2-3 weeks later	Promotion system picks up category + homepage placements based on soft data
Post-release	ongoing	Update waves on the roadmap below; thumbnail changes only with substantial content updates per policy

Table 15.1 - Poki pipeline with owner actions per stage (timelines from official documentation)

15.2 Soft-Release Watchlist

- Conversion to play (C2P) - driven by load time and first-frame quality; target ≥ 65% of clickers reaching gameplay.
- Engagement: median session ≥ 3 minutes; level-3 clear rate ≥ 70% (the tutorial-completion proxy).
- Monetization: ≥ 1 ad per daily active user with a healthy midroll/rewarded split - Poki's stated healthy target.
- Errors: dashboard error section checked twice daily; any repeated console error is a same-day fix.
- Player feedback patterns: falls that feel unfair (specific levels named repeatedly), control confusion in the first 90 seconds, thumbnail mismatch complaints.

15.3 The Word-of-Mouth Engine

Virality is designed, not hoped for. The game's clip-ability rests on three systems. **Auto-captured fail replays**: the deterministic fixed-timestep simulation plus the input buffer means the last 8 seconds of any run can be re-simulated exactly; the game renders the best fall of a session as a ready-made 9:16 share card (PNG via canvas, saved locally at the player's touch - no external URLs, no uploads, fully CSP-safe) showing the fall grade, height, and bounce count over a clean biome backdrop. **TikTok-native readability**: slow-mo falls, polarity-colored field lines, and the fall-grade card are all legible at thumbnail scale with the sound off, which is how clip content actually gets consumed. **Ghost culture**: friend and regional ghosts (via Poki's AUDS data store - a first-party, pre-approved platform feature) turn every level into a leaderboard-shaped social object; beating a friend's ghost produces its own share card. The strategy is one compounding loop: a funny fail gets shared, a shared card carries the game's name and visual identity, a new player arrives and generates their own fail - the same flywheel that made Drive Mad and Level Devil TikTok phenomena, but with the shareable artifact manufactured by the game itself.

15.4 Year-One Update Roadmap

Wave	Timing	Content
W1: Quality + ghosts	Week 2-4 post-global	Ghost polish, assist tuning from live data, QoL from feedback, first seasonal cosmetic set
W2: Chute Mode	Month 2-3	Endless falling-climb mode: one life, altitude scoring, weekly seed, regional ghost boards
W3: Biome 6 - The Core	Month 4-5	Story finale biome + 14 levels + ending vignette + core-themed cosmetics
W4: Level Pack 1 + events	Month 6+	14 remixed-gauntlet levels, seasonal themed events (Poki's guidance: themed content lifts engagement)

Table 15.2 - Post-launch content waves (shared at submission - the room-to-grow signal Poki looks for)

16 Production Plan

16.1 Milestones

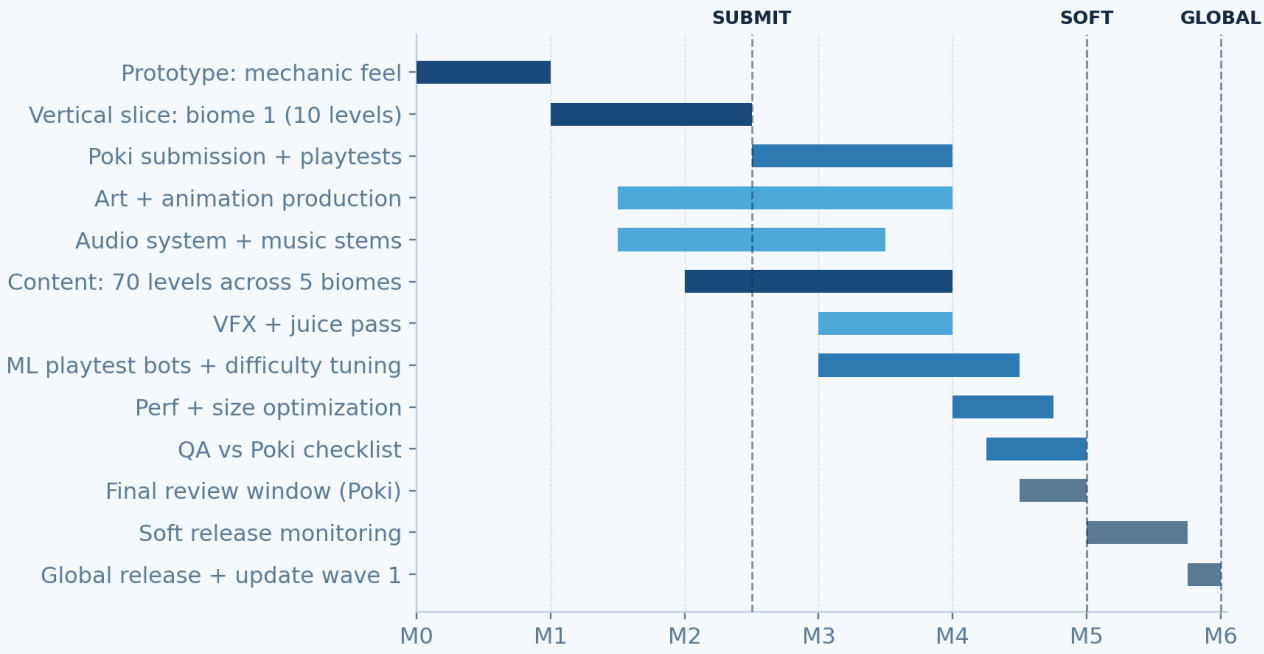


Figure 16.1 - Six-month solo production plan with submission, soft, and global gates

Milestone	Exit criteria
M0 (month 1): Prototype	Magnet controller passes the 10-minute stranger test: fun with placeholder cubes; 20 s loop; fall comedy already emergent
M1 (month 2): Vertical slice	10 polished levels of biome 1; full loop (load→clear→reward); 3.9 MB budget held; ML bot verifies fall-rate band on all 10
M2 (months 3-4): Content production	All 70 levels grey-boxed, 50+ art-finished; all systems feature-complete; telemetry live in dev builds
M3 (month 4): Polish + playtest	Juice pass complete (VFX/audio/motion); 5 external playtests done; Poki Playtesting round 1 submitted
M4 (month 5): Optimization + submission	Performance contract green on full matrix; 3.85 MB payload; QA zero-findings; submit for web fit
M5 (month 6): Soft release	Daily watchlist cadence; difficulty patches from live data; animated thumbnail; A/B static thumbnail
M6 (months 6-7): Global	Promotion pickup; W1 update staged; Chute Mode development begins

Table 16.1 - Milestone exit criteria (a milestone is done when its gate passes, not when its tasks end)

16.2 The Solo + AI Workflow

The weekly rhythm is deliberately craft-forward: Monday and Tuesday are content days (level grey-boxing, prop modeling, hand-balancing), Wednesday is systems day (code, physics tuning, tests), Thursday is juice day (animation, VFX, audio - the differentiators), Friday is verify day (bot sweeps, heatmap review, device matrix, and a 30-minute self-playtest with recording). The AI subscription participates as a pair-programmer and drafting partner under the Chapter 11.4 ledger: code review and refactoring suggestions, test scaffolding, GLSL drafts, localization first passes, and documentation structure. What it never does is ship: every asset, level, and tuning decision passes through the developer's hands, both because Poki's originality bar demands it and because feel - the entire product - is not delegable.

16.3 Zero-Budget Resource Map

- Engine, physics, audio: Three.js, Rapier, Web Audio API - all free and open-source.
- Tools: Blender (modeling/animation/LUTs), Vite + esbuild (build), TypeScript, VS Code, Git/GitHub free tier, GitHub Actions free minutes for CI.
- Audio: self-recorded foley where synthesis needs a base (kitchen junk is literally the theme), Audacity for edits, everything else synthesized at runtime.
- Testing: friends-and-family pool, then Poki Playtesting (free, real players), then Poki's own device telemetry.
- Cash cost of the entire production: the existing Claude subscription and electricity.

16.4 Risk Register

Risk	L×I	Mitigation
Magnet feel never reaches fun	M×H	M0 is a feel gate with a stranger test; 3 fallback tuning presets; pivot option to Defold stack preserved for 60 days
Size budget regression	M×H	CI size gate fails builds over allocation; vertex-color-first art strategy; audio is synthesized
Rejection on originality grounds	L×H	Clone-defense matrix + AI ledger maintained from day one; unique mechanic verified against live catalog monthly
Low-end performance fails	M×M	Reference device in the nightly matrix; dynamic resolution + particle ceilings designed in, not bolted on
Solo burnout / scope creep	M×H	Must/should/could scope ladder; milestone exit criteria kill zombie tasks; 70-level scope is the ceiling, not the floor
Poki pipeline delays	M×M	Submission at M4 leaves slack before the 6-month plan's soft window; content continues during review
Soft-release metrics miss targets	M×H	Watchlist triggers pre-planned interventions: difficulty patches, thumbnail A/B, onboarding trim - not panic redesign
Art inconsistency across biomes	M×M	Style tests automated at cook time; single palette contract; fortnightly silhouette review
Monetization below 1 ad/DAU	M×M	Placement map reviewed against live data; Second Wind offer timing tuned on the fall-grade card
Ghost system abuse (AUDS)	L×M	Server-side validity window; ghost replay must re-simulate within tolerance to be listed

Table 16.2 - Top risks with mitigations (reviewed at every milestone gate)

17 Success Metrics & KPIs

17.1 Stage Gates

Phase	Metric	Target	Action if missed
Playtest (M3-M4)	Stranger 10-min fun verdict	≥ 4/5 testers choose to keep playing	Return to feel tuning; do not submit yet
Playtest	First-clear of level 1	≥ 85% within 90 s	Trim opening; strengthen ghost-hand tutorial
Soft release	C2P	≥ 65%	Loading/first-frame audit; thumbnail A/B
Soft release	Median session	≥ 3 min	Level-length audit vs 90 s budget; reward pacing

Phase	Metric	Target	Action if missed
Soft release	Ads per DAU	≥ 1.0	Placement-map review vs live event data
Soft release	Level-3 clear rate	≥ 70%	Onboarding friction review from recordings
Global	Error rate	Near-zero on dashboard	Same-day hotfix policy
Global	Update engagement	Each wave lifts DAU ≥ 15%	Re-theme waves; deepen ghost features

Table 17.1 - KPI framework by phase (targets set from Poki guidance and documented web-game benchmarks)

17.2 Decision Rules

Numbers exist to make decisions mechanical, and three rules govern the live game. If a level's live fall rate drifts outside its contract band by more than 8 points for three days, it is patched in the next update window - the data outranks the designer's affection. If soft-release C2P misses by more than 15 points after the loading audit and thumbnail A/B, the problem is the first 60 seconds, and the intervention is onboarding surgery, never feature addition. And if the game passes soft release but engagement holds flat through two update waves, the year-one roadmap's fourth slot converts from content to a Chute-Mode early pull - because on a curated platform, the graveyard is full of games that stopped being interesting, and none of them failed their QA checklist.

18 Appendix

18.1 Appendix A: Poki Compliance Matrix (Sign-Off)

#	Requirement	Status
1	Cross-device input (touch/keyboard/mouse-only) + tablet forced touch UI	By design (Ch. 12)
2	16:9 canonical scaling, full-screen coverage	By design (Ch. 10.6)
3	Incognito playable, storage guarded	By design (Ch. 10.6)
4	Zero external requests; bundled fonts/assets	By design (Ch. 10.3)
5	No splash screens, outbound links, foreign ads, ad-block detection	By design (Ch. 2.2 / 14.1)
6	SDK events per spec, CI-verified	By design (Ch. 10.5)
7	Clean production build	Build-mode audit (Ch. 14.1)
8	Save with fallback + notice	By design (Ch. 10.6)
9	≤ 5 MB initial / ≤ 8 MB total; < 10 s playable	3.85 / ≤ 7 MB; CI-gated (Ch. 10.3)

#	Requirement	Status
10	Static + animated thumbnails to spec	628×628 + 3 s loop (Ch. 6.6)
11	Content safety: wholesome, zero gore, no chat, no PII	By design (Ch. 3.3 / 5.6 / 11.5)
12	Originality dossier: clone-defense + AI ledger	Tables 3.2 + Ch. 11.4
13	Session-0 doctrine implemented	Ch. 5.7
14	Localization: 7 launch locales, getLanguage + fallback	Ch. 12.3
15	Update roadmap shared at submission	Table 15.2

Table 18.1 - One-page pre-submission sign-off (owner initials per row at M4 gate)

18.2 Appendix B: Glossary

Term	Definition
AUDS	Poki's Arbitrary User Data Store - first-party key-value storage used here for ghost exchange
C2P	Click-to-play: share of players who start playing after clicking the game card
Chute Mode	Planned endless one-life climb-fall mode (post-launch wave 2)
commercialBreak()	Poki SDK call signaling a natural midroll ad opportunity
DAU	Daily active users
Field lines	Signature VFX: animated dipole ribbons visualizing MAGNO's polarity
Fall rate	Median share of level attempts ending in a fall - the difficulty metric
Polarity flip	The core mechanic: reversing attract/repel to stick, launch, and snap
Second Wind	Rewarded-video fall insurance offered on the fall-grade card
Session 0	The first session - which on web carries most of a game's lifetime value
Snap	Airborne attract-pull to the nearest ferromagnetic surface in cone/radius
Web fit test	Poki's technical/quality test phase before final review

Table 18.2 - Terms used in this document

18.3 Appendix C: Research Sources

Primary platform documentation (fetched and archived October 2026): Poki Developer Guide pages - What We Look For; Requirements; Content & Player Safety (including the clone-assessment dimensions and the Working-with-AI guidance); External Resources Policy; Choosing Your Web Game Engine; Easy Access & Onboarding; Engagement; Localization; Game Thumbnail; How Monetization Works; Monetization Tips; How We Partner With You (deal types); Final Review; Release Process - all at developers.poki.com. Market data: The Business Research Company browser-games sizing (7.81B/2025, 8.01B/2026, 9.07B/2030 forecast); Poki

2026 State of Web Gaming study coverage via GamesBeat and PocketGamer.biz (2,000 players + 400 developers surveyed; 92% positive HTML5 sentiment; 27% spending $50+; 53% of devs porting to web); HTML5 games market forecasts to 2035. Web performance benchmarks: Three.js vs Unity WebGL load-time comparisons (2026); Rapier WASM physics performance discussions; WebGPU browser rollout status (November 2025 milestone). Rejection-pattern context: AI shovelware coverage on Steam (Kotaku on Next Fest, June 2026; aiandgames.com platform-policy analysis; Valve AI-asset rejection reporting). Landscape scans: Poki.com live popular lists (October 2026); title-collision searches for MAGNETOPE and magnet-mechanic games across Poki, CrazyGames, itch.io, and general web search.